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The Negligible Effect of Free Contraception on Fertility: Experimental Evidence from Burkina Faso*

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Abstract

We conducted a randomized trial among 14,545 households in rural Burkina Faso to test the oft-cited hypothesis that limited access to contraception is an important driver of high fertility rates in West Africa. We do not find support for this hypothesis. Women who were given free access to modern contraception for three years did not have lower birth rates; we can reject even modest effects. We cross-randomized additional interventions to address inefficiencies that might depress demand for free contraception, specifically misperceptions about the child mortality rate and social norms. Free contraception did not significantly influence fertility even in combination with these interventions.

KEYWORDS: FAMILY PLANNING; DEMOGRAPHIC TRANSITION; SOCIAL NORMS; RANDOMIZED TRIAL

JEL CODES: J13; J18; O12

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1 Introduction

Over the past 50 years, the number of children born per woman has declined significantly, from a global average of 5 in 1970 to 2.3 today. However, many countries in Sub-Saharan Africa have not experienced this fertility transition to levels near the replacement rate. For example in Burkina Faso, the total fertility rate only declined from 6.7 children in 1970 to 4.8 children today ([World Bank, 2022](#)). While the reasons for this exceptionalism are debated ([Casterline, 2017](#)), arguably the most influential view is that limited access to affordable contraception is the key driver of high fertility in Sub-Saharan Africa ([Bongaarts et al., 1990](#); [Bongaarts, 2017](#)). Under this view, there is a large “unmet need” for contraception: many pregnancies are unintended and could be prevented if women had access to reliable and diverse birth control methods.¹

This view is widely held in policy circles and has shaped government and donor priorities. Over 90% of governments in Sub-Saharan Africa have laws or regulations that guarantee access to contraceptive services ([United Nations, 2021](#)). A global partnership known as FP2030 allows governments and foreign aid agencies to allocate billions of dollars each year in pursuit of universal access to family planning by 2030 ([FP2030, 2021](#)). The goal of these efforts goes beyond giving women control over their reproductive health as a human right. Affordable contraception is often seen as a strategy for economic development, with high fertility viewed as a key barrier to poverty reduction ([Cleland et al., 2006](#); [Ezeh et al., 2012](#); [Canning and Schultz, 2012](#)). However, whether unmet need for contraception is actually driving fertility levels in Sub-Saharan Africa is unsettled in the academic literature, which has long debated the relative importance of access to contraception in determining fertility ([Pritchett, 1994](#)).

We test the hypothesis that the cost of contraception is an important determinant of fertility by conducting a large-scale randomized controlled trial (RCT) in Burkina Faso between 2018 and 2021. In collaboration with the Ministry of Health, we randomly offer women of reproductive age in rural communities vouchers that cover 100% (“full subsidy”) of the cost of contraception at their local government health clinic. In the remaining communities (“control” or comparison group), households are provided vouchers for a 10% subsidy. Our estimates should therefore be interpreted as the effect of offering a full subsidy versus a small subsidy. In

¹This view is supported by the fact that 20 to 40% of women in low-income countries report not wanting to get pregnant in the near future, yet they are not using any modern contraception method—this is the definition of “unmet need” ([UNFPA, 2016](#)).

short, the full subsidy intervention ensures free, local access to modern contraception for three years.

In the spring of 2018, we recruited a baseline sample of more than 14,000 households from 499 villages that cover half of the Burkina Faso provinces. To assess impacts on fertility behaviors, we conducted an endline survey three years later, in the spring of 2021. Despite challenges (a rise in terrorism in many parts of the country and the COVID pandemic), we were able to survey 88% of the baseline sample. The primary outcome we study is realized fertility, but we also analyze effects on contraceptive use.

We find that the full subsidy intervention had no significant effect on fertility during the three-year period, or on the probability and duration of modern contraceptive use, relative to the 10% subsidy comparison group. Based on the 95% confidence interval, we can rule out declines larger than 4.1 percentage points in the likelihood of giving birth over the three-year period, which represents 6.5% of the mean of 62%.² Our large sample size allows us to investigate heterogeneity along many dimensions and to show that the effect remains negligible in many sub-samples, including those that *ex ante* would be expected to have larger unmet demand for modern contraception, such as women who were not using contraception, were not pregnant, and did not want to get pregnant in the near future, and those that face low non-price barriers such as distance to the clinic.

The insignificant effects on fertility and contraceptive use contrast with the significant positive effect of the subsidy on voucher take-up: women who received the 100% voucher were 20% more likely to redeem the voucher than those who received the 10% voucher. Using this measure of contraceptive take-up, free access to contraception would appear to have an effect. However, when we measure contraceptive use in general, accessed through our vouchers and also elsewhere, we find that the subsidy did not induce new users to take up modern contraception. Instead, the vouchers simply paid for contraception for women who would also have used it with only a 10% subsidy.

These precise null effects on contraceptive use and fertility imply that financial constraints are not a first-order barrier to contraception and not an important driver of high fertility in this context. Consistent with this, the women in our sample state that they would like to have six children, on average, *higher* than their realized fertility at baseline. Nevertheless, significant

²The study was powered to detect a decrease in fertility of 3.4 percentage points or larger, based on an *ex post* power calculation that uses the estimated standard error but not the estimated coefficient.

attention and resources are currently devoted to relaxing supply-side constraints. The Burkina Faso government has recently started a national program providing free contraception. (The policy, announced in July 2020 and rolled out in the following months, could not affect births during our study period.) Our results suggest that this program will have little impact on birth rates, at least in the short to medium term. Thus, as argued by other scholars such as [Senderowicz and Valley \(2023\)](#), the rationale for providing free access should be based on arguments regarding reproductive freedom and universal access to basic care, rather than on its impact on birth rates or even birth spacing.

Our finding of low unmet demand for contraception is consistent with a high desired fertility. Understanding whether high desired fertility is an “error” from the couple’s perspective is crucial for understanding the ethical and distributional implications of policies that aim to reduce desired fertility. We shed light on this question through two cross-randomized ‘demand-side’ interventions designed to address potential errors stemming from imperfect information. This allows us to test whether access to free contraception influences fertility once other potential frictions are addressed.

The previous literature in economics has recognized that there are many non-financial barriers to contraceptive use, including misperception of side effects such as fear of infertility ([Bau et al., 2024](#); [Glennerster et al., 2021](#)), misperception of female mortality risks among men ([Ashraf et al., 2022](#)) and disagreement within couples ([Ashraf et al., 2014](#)). We conducted extensive pilot work to investigate demand-side frictions in our context. Although the frictions studied by the literature are also present in rural areas of Burkina Faso, we focused on two dimensions that appeared relevant and have been mostly explored outside of economics: the role of social norms and the role of perceptions of child mortality. We designed two interventions to assess if these frictions could be addressed and whether they impact the price elasticity of the demand for contraception.

Our first demand-side intervention aims to correct potential misperceptions regarding social norms by organizing or showing a film of public debates around fertility and contraception in a random subset of villages. Prior work suggests that social norms influence people’s fertility choices ([Munshi and Myaux, 2006](#); [De Silva and Tenreyro, 2020](#)). Our theory of change was that the debates might change individual beliefs or community norms. First, because the debates present both sides of an issue, they expose attendees to arguments that challenge

their current beliefs.³ Second, the debates might spark ongoing dialogue in the community and shift the social norm. Public discussion might be particularly valuable in contexts like rural Burkina Faso where norms prevent dialogue across generations and genders.⁴ Third, the debates could help correct “pluralistic ignorance,” whereby members of a group privately disagree with a norm but incorrectly assume that all others agree, and the norm persists because of conformism (Allport, 1924; Bursztyn et al., 2020). We created common knowledge about the community views by having attendees vote anonymously, after each debate, on which side represented their view, with the aggregate votes tallied and shared publicly at the end of the meeting. Vogt et al. (2016) show that it is possible to change the acceptability of female genital mutilation by screening filmed debates where actors discuss various positions commonly held in the community. Previous field interventions also suggest that community dialogues can be successful in changing views about contraception in various contexts including Bangladesh (Kincaid, 2000), India (Daniel et al., 2008), Kenya (Wegs et al., 2016) and Niger (Erhardt-Ohren et al., 2023).

Our second demand-side intervention corrects potential misperceptions about the child mortality rate. It is based on the prominent theory in the demography literature that when mortality rates are high, couples adopt a strategy of “hoarding”: they have many children to increase the chances of having a few survive (Notestein, 1953; Davis, 1963; Coale, 1986a; Preston, 1978; Carey and Lopreato, 1995; Kalemli-Ozcan, 2003). Child mortality has halved in Burkina Faso in recent years, going from 179 deaths per 1,000 live births in 2000 to 91 in 2018 (United Nations Inter-agency Group for Child Mortality Estimation, 2024). If high child mortality were a significant driver of high fertility, fertility rates would fall. However, it takes time for individuals to become aware of the new mortality rate. If individuals overestimate the mortality rate (because their beliefs are based on past rates), they might have more children than they would deem ideal. We inform a random subset of households about the level and trend in child mortality in their region of Burkina Faso, using an infographic. The true rate in the infographic is based on data we collected in the same communities, before our baseline survey, in which we administered a birth history module to about 30,000 women covering roughly 200,000 births.

³A number of studies document that individuals tend to expose themselves to individuals who already share similar views, resulting in poor diffusion of information and persistence of shared values – see Levy and Razin (2019)’s review.

⁴A conjecture that motivated this intervention is that social change may be especially slow in societies in which the elderly hold significant power, as is the case in most of West Africa, because it is difficult for younger individuals to express and act on their preferences in such a context.

We find a null effect for both types of demand-side interventions: they had no significant direct effect on fertility or contraceptive use, and they did not modify the effect of the full subsidy. One explanation is that misperceptions are hard to change in a sustained way, and we indeed fail to detect an effect of the interventions on beliefs (whether first-order or second-order) three years later. Complementarities may exist between subsidies and more intensive demand-side interventions, though. An alternative hypothesis consistent with our findings is that, in this economic and social environment, families may well be making a *rational* choice to have large families.⁵

This paper contributes to the literature on the impact of access to contraception on fertility, and in particular on the role of financial constraints. There are a large number of studies including RCTs that have investigated the effects of price on contraceptive use. Two recent reviews conclude that the estimated effect of user fees on contraceptive use is inconsistent across studies and call for more research, particularly to better understand effects on different outcomes (Korachais et al., 2016; Bellows et al., 2016). Surprisingly, virtually all previous experimental studies mentioned in these review articles focus on the effect of contraception access on contraceptive use, but not on fertility. Some notable exceptions are the Matlab experiment in Bangladesh (e.g., Joshi and Schultz (2013)), the Navrongo experiment in Ghana (Phillips and Binka, 2006), and a recent RCT in Malawi (Karra et al., 2022), all of which found that family planning programs significantly reduced fertility.⁶ Both the Matlab and the Navrongo experiments have been criticized for methodological flaws (Miller and Babiarz, 2016). The Malawi experiment only studies women who recently gave birth, so its findings might reflect increases in birth spacing rather than decreases in lifetime fertility. Most importantly, all three experiments included education and counseling, bundling better access to contraception with efforts to increase the demand for contraception.⁷ In contrast, a study in Ethiopia that only provided pills and condoms on credit for three years found no significant effect on fertility (Desai and Tarozi, 2011). Similarly, an RCT in Lusaka, Zambia also found null effects of free provision, though the program only covered injectables and implants, and was only one month

⁵Previous research in West Africa has shown that parents view having many children as offering economic security—to mitigate their lack of savings, insurance, social safety nets, and property rights, especially for women (Rossi and Godard, 2022; Lambert and Rossi, 2016; Donald et al., 2024).

⁶The Matlab intervention lowered lifetime fertility by 14% to 23%, and the Navrongo intervention reduced fertility by 15% (see the review by Miller and Babiarz (2016)). The intervention in Malawi lowered the birth rate from 9% to 5% over a two-year period.

⁷For example the Matlab program offered family planning, reproductive health, and child health services, see Joshi and Schultz (2013).

long ([Ashraf et al., 2013](#)).⁸

We build on this literature to show that making all contraceptive products completely free for a *sustained period* for all women of childbearing age in rural areas (where most of the population of West Africa lives) is insufficient to affect fertility—even in the presence of interventions targeting potential misperceptions, which have not been tested before. Our experiment has sufficient statistical power to detect even modest changes in fertility over a meaningful period of three years.

2 Study Design

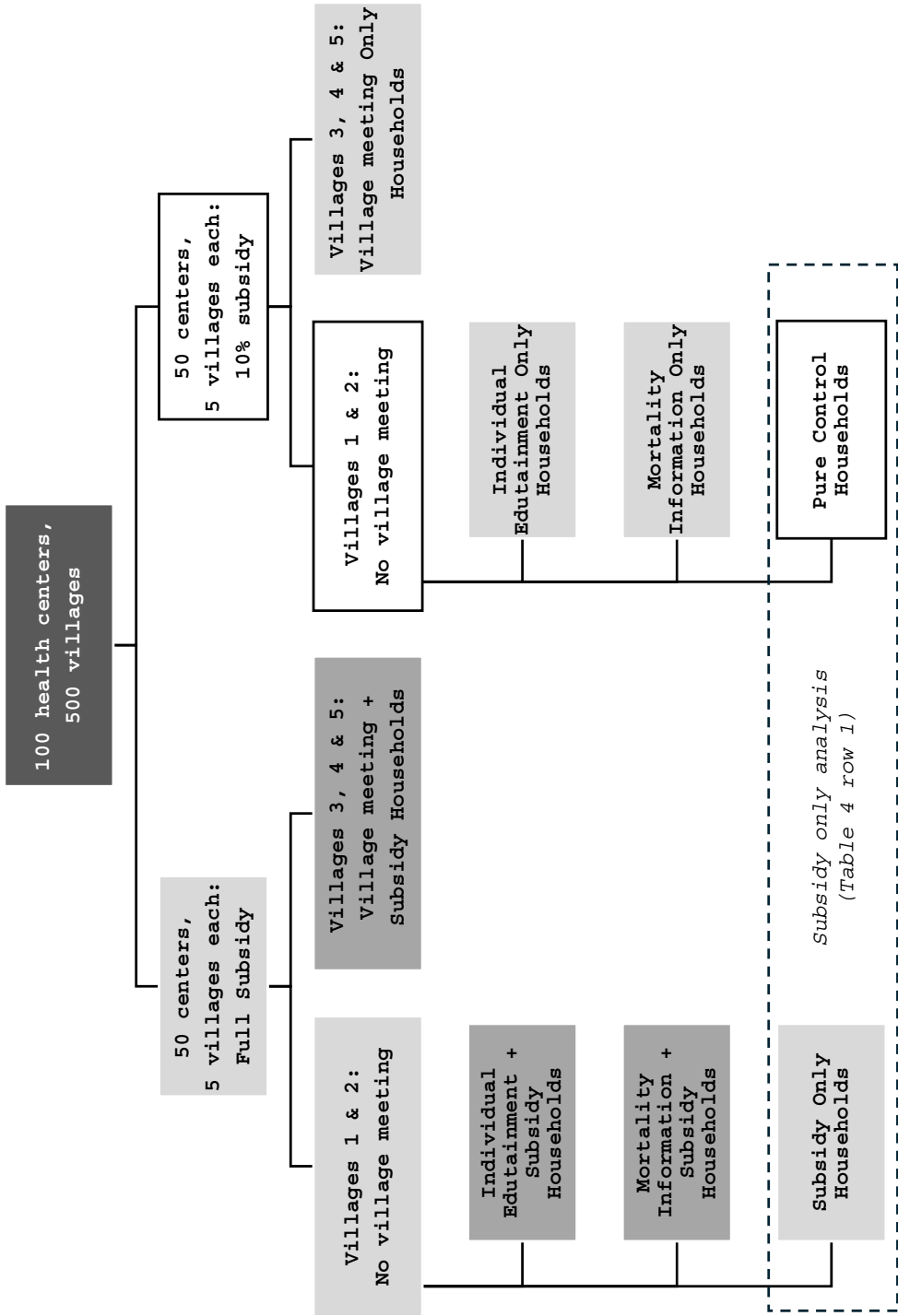
2.1 Experimental Design

[Figure 1](#) presents the experimental design. Our sampling frame starts with 100 public health centers across 20 provinces, sampled from the total of 1,500 health centers in the 45 provinces of Burkina Faso (see map in [Figure A.1](#)). We sampled 5 villages (from among the roughly 10 villages) in the catchment area of each of the 100 health centers, for a total of 500 villages. Health centers were sampled so as to ensure a minimum of 12.5 kilometers between any two health centers in the sample; villages in the catchment area of health centers were selected to maximize the minimum distance between any two villages sampled.

The first randomization was done across health centers: 50 health centers (serving 250 sample villages) were selected for the vouchers for free contraceptives intervention. Women in the other 50 health centers received a 10% discount voucher. This allows us to compare administrative voucher redemption rates across the two arms. We did not expect the 10% discount to have a large effect on contraception use, so this was meant to approximate the status quo.

⁸Using the same experimental data and focusing on the treatment arm with free access, [Ashraf et al. \(2014\)](#) show that the way in which vouchers are delivered matters: involving husbands reduces take-up, with subsequent effects on fertility, compared to giving vouchers to women in private.

Figure 1: Experimental design



Notes: The figure presents the experimental design. There are three levels of randomization. (1) Across health centers: 50 centers (encompassing 250 villages) were assigned to the full subsidy arm and 50 centers were assigned to the 10% subsidy arm. (2) Across villages, within each health center catchment area: 3 villages were assigned to village meetings and 2 villages were assigned to individual interventions. (3) Across households, within each village allocated to individual interventions: some were assigned to the individual edutainment arm, some households were assigned to the mortality information arm, and the rest received no demand intervention. Cells colored in dark gray received both the full subsidy and a demand intervention. Cells colored in light gray received either the full subsidy or a demand intervention. The white cells received neither the subsidy nor any demand intervention: these are the pure control households. See more details in [Figure A.2](#) in Appendix.

The second randomization was done across villages within each health center catchment area. In each catchment area, we allocated 3 villages to receive a group intervention and 2 villages to individual interventions. The group (i.e., village-level) intervention entailed a village meeting with debates or a viewing of an edutainment film on the debate topics. The third randomization was across households within each village allocated to individual interventions. In each village, households either (i) received information about child mortality, (ii) watched the edutainment film at home, or (iii) did nothing.⁹

The goals of the design were to (i) estimate the effect of the subsidy, (ii) estimate the effect of the subsidy when paired with the demand-side interventions, and (iii) estimate the effect of the demand-side interventions on their own. In the next sections, we provide details on the supply-side intervention (the full subsidy) and then describe our household sampling and data collection strategy. The demand-side interventions are described in detail in [subsection 5.2](#).

2.2 Intervention: 3-year-long voucher for free contraception

This intervention, approved by the Ministry of Health in Burkina Faso, was implemented by Innovations for Poverty Action (IPA) in partnership with government health centers. The implementation was identical across the two subsidy levels (100% and 10%).¹⁰

Vouchers were delivered to all eligible women in sampled households, in private, during the baseline survey in spring 2018. The vouchers were accepted by 70% of women; a quite large share (30%) rejected them out of hand.¹¹ The value of the voucher is small by high-income context standards. In our baseline survey, health centers reported charging on average 2.5 US dollars (USD) to insert an implant and USD 2.7 to remove it (implants are the widely used contraceptive product); these prices range between USD 1 and 6.5. However, this monetary cost is large when compared to local incomes: in rural areas, half of the population lives on USD 250 a year or less ([Institut National de la Statistique et de la Demographie, 2019](#)).

⁹See [Figure A.2](#) for more details.

¹⁰A memorandum of understanding was signed between IPA and each health center. IPA committed to cover 100%/10% of the value of family planning (FP) services rendered to women from the 5 villages in the health center's catchment area sampled for the study. In addition, IPA provided a financial compensation to the health center's agents trained to implement the program (identifying eligible women and recording their visits in a dedicated register). IPA made an initial payment at the onset of the program to ensure that health centers did not need to pay in advance for the cost of the subsidized contraceptives and did not run out of stock.

¹¹Among those who refused, the majority said that they did not need contraception; another 20% said they did not want the voucher because they feared the husband or relatives of the husband could see her with the voucher; and 5% mention fear of side effects.

All family planning (FP) services, including consultation for side effects or removal of implants and intrauterine devices, were eligible for the discount.¹² In the 100% arm, all costs were covered, not only contraceptive products but also ancillary products like cotton, alcohol, etc. Records kept by health centers during the intervention show that the most common contraceptives sought were implants, then injections, and to a much smaller extent, oral pills. Stock-out rates for these items were below 5% during monitoring visits, and health centers reported that they could typically restock within a few days.

There are three important points regarding the implementation of the intervention. First, the study was initially planned to last for 2 years, until the spring of 2020. However, we had to postpone the collection of the endline data by one year due to the COVID-19 pandemic. We therefore extended the coverage of the vouchers from 2 to 3 years.¹³ In July of 2020, the government of Burkina Faso announced it would launch a nationwide free FP policy. We decided to keep our program running because it was not clear that the policy would be immediately implemented in all centers and we wanted to ensure continuous free access in the treatment group.¹⁴ Second, Burkina Faso experienced a strike among health workers from June to December 2019. Nonetheless, the strike only affected reporting activities to the Ministry of Health, but not the provision of services. In our monitoring surveys, 92% of our health centers reported that all services were fully available to patients during the strike. Third, four health centers in our sample were permanently closed in June 2019 due to insecurity issues.¹⁵ Our coefficients should therefore be interpreted as intent-to-treat estimates. We discuss the implications of these macro shocks for our findings in Online Appendix C.

¹²If women opted for long-lasting methods, they could visit the health center to interrupt them whenever they wanted (potentially, just before the expiration date of the voucher); these costs were covered.

¹³We informed health agents of this change through in-person visits to health facilities, and health agents relayed the information to women in the village.

¹⁴Figure A.3 report the dates when the free FP policy was implemented in our health centers according to the endline survey. The policy was piloted in the second semester of 2019 in two regions. In a robustness test, we check that our results remain the same if we exclude these regions. Most centers implemented free access during the second semester of 2020, meaning that the policy could only affect births at the very end of our study period (endline data was collected between February and June 2021). Finally, some health centers did not start implementing the free FP policy until 2021, justifying our choice to extend the duration of our program.

¹⁵Burkina Faso experienced a major rise in jihadist terrorism threats over our study period, as shown in Figure A.4.

2.3 Sampling of households and randomization

To enroll households in the study, we used the same protocol across all 500 villages. We started by doing a “census” listing to identify the eligible households. The criterion to be eligible for the study was that the household had at least one woman who was (i) between the ages of 17 and 35 years old, (ii) married and living with her husband, and (iii) neither pregnant nor the mother of a child under age 6 months at the time of the survey.¹⁶ In households with multiple eligible women, we randomly chose one as the “focal” woman. If she was in a polygamous marriage, information on co-wives was collected through the husband.

Our sample is representative of a large share of the population in Burkina Faso. According to the 2019 census, 74% of the national population lives in rural areas. The average woman in our sample is comparable to the average woman in rural areas nationwide, having similar education, religion, polygamy status and household assets ([Institut National de la Statistique et de la Demographie, 2022](#)).¹⁷ Low urbanization rates are not specific to Burkina Faso. Overall, 54% of the population in West Africa live in rural areas; the fraction ranges between 60% and 85% in several countries, such as Guinea, Niger, Sierra Leone and Togo ([United Nations, 2018](#)).

We randomized health centers to free FP services, stratifying by province, and randomized villages to group interventions, stratifying by health center. In each village assigned to a group intervention, we randomly sampled 25 eligible households. In each village assigned to individual interventions, we randomly sampled 35 households and randomized them to one of the individual treatments (mortality information or private screening of the edutainment video, described in [section 5](#)), stratifying by village. We sampled more households in villages with individual interventions to increase the statistical power of the tests comparing treated and control households within a village.

To ensure balance, our randomization procedure followed [Imbens \(2011\)](#). We generated thousands of potential random assignments. For each assignment, we tested for balance across the different arms we intended to compare, for a set of key variables collected during the base-line listing. We set criteria for a potential assignment to be considered balanced, and then

¹⁶We excluded women who had recently given birth since the locally recommended FP method for them at the time was exclusive breastfeeding.

¹⁷In 2019, 75% of women in rural areas had no formal education, 64% were Muslims, 38% had a polygamous husband, 4% of rural dwellings relied on the national electricity grid to illuminate their homes, 50% had a radio, 54% had a toilet, 55% had a cemented or tiled floor.

we randomly chose one of those balanced randomizations as the final assignment.¹⁸ [Table A.1](#) shows that the procedure was successful in creating balanced groups in terms of fertility preferences, contraceptive use, important demographics and potential sources of heterogeneity at baseline.

3 Data

3.1 Data collection

Listing. The census listing took place in Fall 2017. In addition to socio-demographic information on 68,241 households in our 500 study villages, it contains survival data for 190,706 births across the period 1973-2012. Given its size and the paucity of mortality data, our listing data is itself a contribution. It reveals important sub-national heterogeneity in both levels and trends, awareness of which enables a government to appropriately target resources ([Figure A.5](#)).

Baseline. We conducted a baseline survey with the focal wife in the spring of 2018 in 499 villages; one village could not be surveyed due to security issues. We completed baseline surveys with 14,545 focal wives, who are our primary respondents, as well as shorter surveys with 10,683 husbands.¹⁹ Each spouse was surveyed in private. The outcome measures collected in the survey were fertility history (pregnancy and births) and proximate determinants of fertility (desired family size, contraceptive use). We collected two types of secondary outcomes. One type represents intermediate steps in the theory of change: affordability of contraceptives, perceptions about child mortality, and community norms about fertility and contraception. The other type are measures of physical and overall well-being (e.g. self-reported health status, happiness and overall life satisfaction, spousal communication), since well-being could be affected by the interventions even without changes in fertility (e.g., knowing that FP services are available for free if one ever needs them may improve a woman’s well-being). To measure these variables, we used questions from the Demographic and Health Surveys or newly developed questions that we validated through piloting.

¹⁸More precisely, we generated 300,000 randomizations. We had 10 key variables and 10 pairwise comparisons (between the 5 treatment arms at the village level), so 100 comparisons in total. We defined a randomization as balanced if all 100 comparisons showed a standardized difference between groups below 0.2. This was the case for 2,232 randomizations.

¹⁹Roughly one quarter of the husbands were away from the village during the home visits and could not be surveyed. This fraction is the same in all treatment arms (see [Table A.2](#), column 4).

Endline. Due to COVID-19, the endline survey did not take place in 2020 as planned, but in the spring of 2021, three years after the baseline. The security situation in Burkina Faso had deteriorated since the summer of 2018, preventing free movement in the national territory. Therefore, the endline survey could not be conducted in person in all villages. We had to conduct the survey by phone in 34% of the villages (see [Figure A.6](#)). The phone version of the survey was shorter than the in-person survey and focused on primary and secondary outcomes. Fortunately, the stratification by province, health center and village ensures that all treatment arms are equally affected by these issues, which are highly spatially correlated.

We were able to survey 87% of women and 88% of men initially sampled, which is very high given the disruptions that occurred in this time period. Attrition is higher in places surveyed by phone: 18% compared to 10% in places surveyed in person. Attrition is particularly concerning if it is differential. [Table A.3](#) shows that the attrition rate is balanced across the full and 10% subsidy groups.

Monitoring and administrative records. Once a year, we visited the health centers to collect (i) voucher redemption data and (ii) monitoring data on stocks of contraceptives, prices charged, and health-worker strikes. In 2019, we sampled a few households from each village and surveyed them about their health center visits in order to audit the quality of the administrative records. Health centers were found to be following the study protocol.

3.2 Baseline summary statistics

[Table 1](#) reports summary statistics for the focal wives at baseline.

The average woman is around 28 years of age and is married to a 38 year-old man. In this agricultural setting, most women work in the fields of the family farm. 83% of women have no formal education and 45% of women live in polygamous households. Roughly two thirds are Muslim. Households are poor: for example, only 1% have access to an electricity network, 48% own a radio, 47% of the dwellings have cemented or tiled floor and 43% have a toilet.^{[20](#)}

The average women had 3.5 pregnancies before baseline. Desired fertility is high. Almost all women (92%) want another child, and 35% want a child in the next 2 years. Women report wanting a total of 6 children. There is little variation in the ideal number of children: half of

²⁰The sample is not constant due to attrition or non-response for outcomes. Some baseline characteristics are also missing.

Table 1: Summary statistics from baseline survey of wives

	Mean	SD	N
Wife's age	28.24	5.45	14,607
Husband's age	38.26	11.54	14,051
Wife reports husband is polygamous	0.45	0.50	14,609
Muslim	0.64	0.48	14,597
Wife has no formal education	0.83	0.37	14,605
HH has access to electricity network	0.01	0.12	14,607
HH has a radio	0.48	0.50	14,603
HH has a toilet	0.43	0.49	10,022
HH has a cement/tiled floor	0.47	0.50	9,920
Fertility:			
# of pregnancies before baseline	3.51	1.84	12,543
Wants another child	0.92	0.27	14,609
Wants another child in next 2 years	0.35	0.48	13,931
Total # of children desired	6.00	1.87	13,212
Exposure to contraception:			
Ever heard of contraception/ methods to delay births	0.91	0.28	14,602
Ever used modern contraception	0.46	0.50	14,595
Currently using modern contraception	0.31	0.46	14,590
Share of implant users (lasting for 3 to 5 years) among current users	0.59	0.49	4,589
Share of injectable users (lasting for several months) among current users	0.32	0.47	4,589
Distance to local health center (kilometres)	6.20	4.57	14,609
Has unmet need for contraception	0.38	0.49	14,596
Could not afford contraception if ever wanted to use it	0.41	0.49	13,240

Notes: Data from Baseline survey with wives. See Table A.4 for husband data. As standard in the literature, a woman is considered as having unmet needs for contraception if (i) she is not currently using contraception and (ii) she does not want another child in the next two years.

women want 5 or 6.

Almost all women know about modern contraception. 46% of women have ever used modern contraceptives and 31% are currently using them at baseline.²¹ Among current users, 59% use implants (lasting for 3 to 5 years) and 32% use injectables (lasting for several months). This relatively low use does not appear to be driven by a rejection of these methods: only 20% of individuals think modern contraception is dangerous to health. Finally, almost 40% satisfy the conditions for having an ‘unmet need for contraception’ (they do not want a child and are not using modern contraception) and 41% report they would not be able to afford contraception if they wanted to use it.

²¹For reference, 65% of U.S. women aged 15–49 were using a contraceptive method in 2018: <https://www.cdc.gov/nchs/data/databriefs/db388-H.pdf>

4 The effects of a full subsidy for contraception

4.1 Regression specification

We estimate the impact of the supply intervention (vouchers for free contraception) with the following model:

$$Y_{ivcp} = \beta_1 FullSubsidy_c + \gamma_p + X_{iv}\rho + \varepsilon_{ivcp} \quad (1)$$

where Y_{ivcp} is the outcome of interest (typically fertility or contraceptive use) for a given individual i in a given village v covered by health center c located in province p . $FullSubsidy_c$ is an indicator variable for whether the health center was assigned to the 100% subsidy. γ_p is a set of province fixed effects. We add them to account for the stratified design. In our baseline specification, the controls in X_{iv} indicate whether the village had to be surveyed by phone due to security concerns and the date of the endline survey (as a continuous variable measured in days). We report heteroskedasticity-robust standard errors clustered at the level of randomization, namely the health center.

As a robustness check, we include a set of individual controls measured at baseline in X_{iv} . Most importantly, we include the baseline outcome (when available) and additional controls that are highly predictive of fertility behaviors in previous work (wife’s age, spousal age gap, polygamous union, whether the wife has ever gone to school, number of births at baseline, whether she has had a child who died, whether the husband has been surveyed, and DMI radio exposure).²² Adding controls should not affect the point estimates (if the randomization worked) but can reduce the variance of the error term and thus improve the precision of the estimates. The results are similar if we include controls.

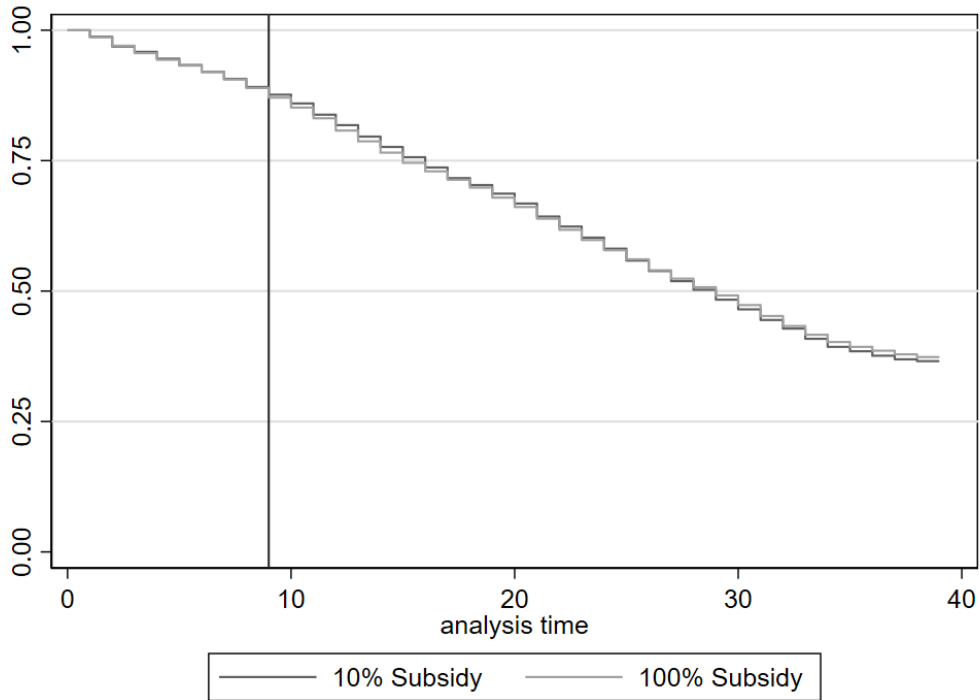
We also estimate the effect of the subsidy in sub-populations that would be expected to have the largest effect of free contraception—those for whom the cost of contraception seems most likely to be a binding constraint. We identify these sub-populations using baseline information, for example on financial constraints or stated unmet need for contraception.

²²The baseline survey collected information on all baseline outcomes. When the values of any controls are missing we impute a value and include an indicator dummy equal to one if the variable was imputed. DMI radio exposure is included to account for the fact that DMI, an international NGO, broadcast a radio show about FP in parts of Burkina Faso during the study period. See [Glennerster et al. \(2021\)](#) for more information.

4.2 Average effects

We start by investigating the effect of the full subsidy on fertility. A preliminary way to summarize the results is to plot the fraction of women who have not given birth to another child as a function of time since the intervention began, measured in months (Figure 2). By construction, the fraction is 100% at $t=0$. The interventions could only influence births taking place at least 9 months (shown as a vertical line) after the intervention start. At the 9-month mark, 12.5% have already given birth suggesting that many women were pregnant but did not know it at the time of the baseline survey and thus were included in the study. Importantly, and this confirms balance across the two arms, there are no differences up to that time between the full subsidy group and the comparison (10% subsidy) group.

Figure 2: Duration until next birth, by subsidy status



The figure shows a Kaplan-Meier Survival curve for the 10% subsidy group and the 100% subsidy group. The y-axis shows the fraction of people who have not given birth, starting from the date of the baseline survey ($t=0$) and up to the endline survey ($t=36$ months). The intervention could not have affected fertility in the first 9 months (area left of the vertical line). The fraction of women who have given birth during the first 9 months, and hence were presumably pregnant at baseline, is equal to 12.5%. The fraction is the same in both groups. A test that the survival curves are identical cannot reject the null ($p\text{-value} = 0.55$), thus the fertility behavior post intervention is the same regardless of the contraception subsidy received.

If the intervention had an effect, we would expect to see fertility diverging after 9 months.

Instead, we observe that the curves follow each other closely, suggesting no impact of the full subsidy voucher. Indeed, a test of equality of the Kaplan-Meier survival curves cannot reject the null that the curves are the same ($p\text{-value} = 0.55$). Moreover, there does not appear to be any timing differences that emerge at any point. Under the hypothesis that free contraception only affected the spacing of birth, but not the probability of birth after 3 years, the slope of the 10% curve would be steeper than the slope of the 100% curve right after the 9-month mark (women without free access would have another birth sooner) but eventually both curves would converge towards the end of the intervention period. This is not what we observe: women in both groups give birth at the exact same rate during the whole period. This result implies that any aggregate measure of fertility will not be affected by the voucher because we do not detect differences in fertility rates at any time during the 3-year duration of the study.

This null result is also seen in column 1 of [Table 2](#), which reports the results from estimating equation 1, without baseline controls in Panel A, and with baseline controls in Panel B. In Panel A, those receiving the full subsidy have a 1.7 percentage point lower probability of having a live birth in the three years since the baseline, which is a 2.7% decrease relative to the control mean of 62%. This is a small and statistically insignificant effect. One explanation for the absence of effect on live births could be that access to contraception leads to fewer and healthier pregnancies. Pregnancies would decrease, but miscarriages would also decrease conditional on pregnancies, which would leave live births unchanged. Since we collected data on pregnancies, we can directly test and rule out this hypothesis: in column 2 of [Table 2](#), we find a 1.9 percentage point lower probability of having a pregnancy, a 2.7% decrease relative to the control mean of 70.5%. The coefficient is insignificant. Therefore the magnitude and precision of the effect are similar for live births and pregnancies.

Table 2: Treatment effects on primary outcomes

	(1)	(2)	(3)	(4)	(5)
	Had a live birth since baseline	Had a pregnancy since baseline	Used medical contracep- tion in last 3 yrs	Month(s) used modern contraception (last spell of each type) in last 3 years	Used IPA subsidy voucher
Panel A: with only endline controls					
Full Subsidy	-0.017 (0.012)	-0.019 (0.012)	0.0003 (0.015)	-0.318 (0.399)	0.032*** (0.010)
Baseline Controls	No	No	No	No	No
Panel B: with endline and baseline controls					
Full Subsidy	-0.017 (0.011)	-0.018 (0.011)	0.008 (0.013)	-0.194 (0.360)	0.035*** (0.009)
Baseline Controls	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes
Observations	12,542	12,543	12,131	12,107	12,519
Control (10% Subsidy) Mean	0.623	0.705	0.531	9.609	0.142

Notes: Endline controls are: Whether the village had to be surveyed by phone due to security concerns at endline and date of the endline survey. Panel B additionally has the following baseline controls: number of births at baseline; whether wife was using modern contraception at baseline; number of children desired by wife at baseline ; number of children desired by husband at baseline; under 5 mortality rate reported by wife at baseline; wife and husband's first-order beliefs at baseline i.e. whether or not they each agree that "there is a quantity-quality trade-off", "times are changing and there is no social norm on family size" and (in response to a vignette) "Z. should use long-lasting contraception to delay 5th birth"; wife's age at baseline, spousal age gap at baseline, polygamous union at baseline, whether husband was surveyed at baseline, whether the wife has ever gone to school, whether she has had a child who died at baseline, exposure to DMI radio programs at baseline. Province fixed effects are used in all specifications. Robust standard errors in parentheses. Clustering at health center level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Consistent with the lack of effects on fertility, the fraction of women using contraception in the last 3 years did not increase in response to the subsidy: receiving a free voucher increases self-reported contraceptive use by 0.03 percentage points, less than a 0.1 percent increase (column 3 of Table 2). There is also no effect on the intensive margin of use: the subsidy led to a statistically insignificant and very modest decline in the reported number of months that women used contraception in the last 3 years (a decrease of 0.3 months relative to a mean of 9.6 in the comparison group, column 4 of Table 2).²³ The point estimates and significance levels are very similar in Panel B, when we add baseline controls.

This null result is important for policy. The 95% confidence interval on having a birth is [-0.041; 0.007]. We can rule out a fertility decline larger than 6.5% ($=0.0406/0.623$). Extrapolating from the three-year effect to the effect of lifetime access, 30 years of free contraception (from age 15 and 45) would lower fertility by at most $0.041 \times 10 = 0.41$ children per woman from a starting point of 6.2 children. While not insignificant, this is a small effect relative to what

²³When we look at the method mix, we find no detectable effect: women do not switch from one contraceptive method to another.

one would expect if financial barriers were the main factor behind high fertility.

Why are households not sensitive to the price of contraceptive services? We know from audits and surveys with female participants that implementation problems can be ruled out, i.e., clinics did honor vouchers. In the endline survey, only 13% of women report not having heard of the IPA vouchers. We also know that voucher use was not zero: 14% of women in the comparison group and 17% of women in the treatment group reported redeeming the voucher at least once (Column 5 of Table 2). This 3 pp difference is significant at the 1% level. In the health centers' registers, the difference is 4 pp and is also significant at the 1% level. The fact that voucher redemption is higher in the treatment group confirms that women understood the discount associated with their voucher, being more likely to use it when it was more generous. However, the fact that we see no increase in contraception use in column 3 suggests that all or most of those who redeemed the free contraception voucher were *inframarginal*: they would have used contraception in any case. Thus, the intervention subsidized the use of modern contraception among individuals who would also have used it with only a 10% subsidy.

The voucher take-up in the treatment group (17%) may seem low compared to the fraction of women currently using contraception at baseline (31%), raising the question of whether *inframarginal* women (those who would have used contraception in any case) left money on the table by not using their voucher. However, recall that 59% of baseline *users* had a long-lasting implant, and thus either needed no replacement during the study period (the implants last 3 to 5 years) or planned to get pregnant after the expiration. The voucher usage rate is higher than the 13% share of women who were using a form of contraception other than implants (mostly injectables) at baseline. In addition, some women may prefer to get contraception from a different provider than the health clinic where their voucher was valid, potentially because of a more convenient location, even though they had to pay for it.²⁴

More generally, we investigate whether free contraception affected women's well-being in Table A.5. We find that the full subsidy treatment seemingly *decreased* the incidence of intimate partner violence (IPV) but had no effect on self-reported health, life satisfaction, various measures of reproductive control, and monogamy.²⁵

²⁴While it is possible women are getting free contraception from other sources (including towards the end of the period from government sources), in the endline survey, 78% of women currently using contraception in the comparison group said they had to pay for it.

²⁵Thus, although the full subsidy did not impact fertility substantially, it improved relations between spouses. This is not due to an improvement in the communication between spouses regarding fertility and contraception: If anything the subsidy decreased the number of times spouses talked about desired fertility and contraceptive

4.3 Effects in sub-populations with the likely largest effects

Although our overall impacts on fertility are small and insignificant, these results could mask heterogeneity. We investigate this by estimating the treatment effects within subgroups that would *a priori* be expected to have higher responses to the subsidy.

We first examine women who have unmet demand for contraception based on their baseline survey responses. We start with the UNFPA definition of unmet need: the woman does not want another child in the next 2 years *and* is not using modern contraception. Then, we look at both dimensions separately. We also look at women whose husband did not want a child in the next 2 years. In addition, we note that, despite it being an exclusion criterion, 12.5% of women were indeed pregnant at baseline (but did not know it), and hence presumably did not need contraception at the beginning of the intervention (although they may have needed it after the birth). We therefore also look at the subset of women who were neither pregnant at baseline nor using modern contraception. Finally, we consider women who stated that they could not afford contraception if they wanted to use it.

The results are shown in [Table 3](#), which reports the same outcomes as [Table 2](#) (see [Table A.6](#) for the specification with baseline controls). Since we are testing a large number of hypothesis among sub-samples, we report q-values correcting for multiple hypothesis testing (MHT) ([Anderson, 2008](#)). The effect sizes are still surprisingly modest. For instance, the point estimates and the standard errors for women who were neither pregnant nor using contraception are very close to the whole sample. The largest effects on fertility that we estimate are for women who could not afford contraception at baseline (2.5 pp, a 4% decline in live births and 3.2 pp, a 4.5% decline in pregnancies). However, the effects of the 100% subsidy on live births and modern contraception use are statistically insignificant for all subgroups, even without MHT corrections. The effects on pregnancies are insignificant once we adjust for MHT (all of [Anderson's](#) q-values in columns 1 to 4 exceed 0.10).

Why do women who do not want to get pregnant not take up modern contraception that is provided for free? One possibility is that pregnancy intentions may be ambivalent—that is women may have unresolved or contradictory feelings about having a child. Ambivalence is

use ([Table B.1](#)). Instead, we hypothesize that the subsidy lowered tensions between spouses related to financial issues—we know that most of those taking up the vouchers were already using contraception but they were likely paying a higher price for it. However, these results are to be taken with a grain of salt: the sample is smaller and we do not have baseline IPV data to verify balance pre-intervention. We do not study divorce – 99 percent of respondents stayed married throughout the study.

Table 3: Treatment effects on primary outcomes by subsamples

		(1)	(2)	(3)	(4)	(5)
Sub-sample		Had a live birth since baseline	Had a pregnancy since baseline	Used medical contraception in last 3 yrs	Month(s) used modern contraception (last spell of each type) in last 3 yrs	Used IPA subsidy voucher
Need for Contraception:						
Had unmet need for contraception at baseline (N=4,649)	Full Subsidy	-0.011 (0.015)	-0.015 (0.014)	0.018 (0.019)	-0.104 (0.459)	0.042*** (0.014)
	q-value	0.571	0.327	0.779	0.948	0.007
	Control Mean	0.662	0.736	0.437	7.229	0.128
Wife did not want another child over next 2 yrs at baseline (N=7,583)	Full Subsidy	-0.016 (0.013)	-0.021 (0.013)	0.009 (0.017)	-0.392 (0.462)	0.037*** (0.013)
	q-value	0.571	0.250	0.862	0.896	0.007
	Control Mean	0.611	0.703	0.559	10.649	0.157
Husband did not want another child over next 2 yrs at baseline (N=4,724)	Full Subsidy	-0.013 (0.016)	-0.036** (0.016)	0.022 (0.019)	0.159 (0.514)	0.033** (0.013)
	q-value	0.571	0.118	0.779	0.948	0.015
	Control Mean	0.612	0.712	0.560	10.594	0.155
Was not using modern contraception at baseline (N=8,191)	Full Subsidy	-0.011 (0.012)	-0.014 (0.012)	-0.002 (0.014)	-0.303 (0.352)	0.029*** (0.010)
	q-value	0.571	0.327	0.947	0.896	0.007
	Control Mean	0.641	0.709	0.421	6.854	0.113
Was not using modern contraception and was not pregnant at baseline (N=7,052)	Full Subsidy	-0.017 (0.014)	-0.017 (0.014)	0.001 (0.015)	-0.040 (0.372)	0.034*** (0.010)
	q-value	0.571	0.311	0.947	0.948	0.005
	Control Mean	0.585	0.677	0.408	6.551	0.109
Could not afford contraception at baseline (N=4,519)	Full Subsidy	-0.025 (0.017)	-0.032** (0.015)	0.008 (0.017)	-0.099 (0.497)	0.037*** (0.013)
	q-value	0.571	0.118	0.862	0.948	0.007
	Control Mean	0.616	0.702	0.505	9.055	0.148
Other Frictions:						
Health Center < 2km away (N=2,060)	Full Subsidy	-0.025 (0.028)	-0.040 (0.029)	0.043 (0.036)	0.827 (0.965)	0.095*** (0.023)
	q-value	0.571	0.304	0.779	0.896	0.001
	Control Mean	0.578	0.680	0.558	10.022	0.150
Disagrees with modern contraception being harmful for health (N=8,563)	Full Subsidy	-0.007 (0.013)	-0.013 (0.014)	-0.007 (0.016)	-0.641 (0.427)	0.036*** (0.011)
	q-value	0.571	0.334	0.862	0.896	0.005
	Control Mean	0.617	0.705	0.573	10.633	0.159
Neither wanted another child over next 2 years (N=3,771)	Full Subsidy	-0.010 (0.018)	-0.037** (0.018)	0.025 (0.021)	0.036 (0.536)	0.035** (0.015)
	q-value	0.571	0.118	0.779	0.948	0.023
	Control Mean	0.599	0.709	0.567	11.044	0.163

Notes: Endline controls are: Whether the village had to be surveyed by phone due to security concerns at endline and date of the endline survey. Province fixed effects are used in all specifications. Robust standard errors in parentheses. Clustering at the health center level. Control Mean refers to that of 10% subsidy sub-sample. N refers to the number of observations in each subsample. The number of observations may vary slightly between outcomes; in this case, we report the minimum number across all 4 regressions. q-value refers to the adjustment for Multiple Hypothesis Testing in [Anderson \(2008\)](#). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

systematically correlated with lower use of contraception in various contexts ([LaCross et al., 2019](#)). In West Africa, qualitative studies document that pregnancy intentions are poorly specified and vary over time for many women ([Agadjanian, 2005](#)). In Burkina Faso, [Speizer \(2006\)](#) assesses the strength of fertility intentions among women. She finds that a quarter of

those who expressed wanting to delay their next birth or to stop having children also reported that a pregnancy in the next few weeks would not be a problem. Our small effect sizes suggest that estimated rates of unmet need for contraception may overestimate the share of women who unambiguously want to avoid getting pregnant. Alternatively, it may be that women can control their fertility using traditional contraception methods. Indeed fertility levels fell in many now-rich countries prior to the availability of modern contraception (Coale, 1986b). Recent research shows that traditional birth control methods are also used effectively today in other parts of Africa (e.g. Alam and Pörtner (2018)).

An alternative explanation could be that women do not want to have children and cannot control their fertility but face other barriers that prevent them from using modern contraception. Our findings suggest that financial constraints are not the main or only barrier to contraceptive use for most women. Either women do not want to use contraception or they face other barriers, and its free provision does not ameliorate these issues.²⁶ To investigate this, we test whether the subsidy had larger effects on contraception use and fertility among subgroups for whom other frictions are small.

We consider three factors besides cost that might deter people from using contraception even when they want to control their fertility. The first is the distance to the health center. While the full subsidy makes modern contraception free, it is well known that many other barriers prevent individuals from using health care, including the time cost to access such services (Karra et al., 2022). The second is the perception (quite commonly cited in Sub-Saharan Africa) that modern contraception has significant negative side effects (Glennerster et al., 2021; Bau et al., 2024). Indeed in our focus groups some cited fears of infertility and other health concerns when asked about why they opted out of modern contraception.²⁷ Again Table 3 shows there are no statistically significant declines in fertility among groups for whom these additional frictions are low. In particular, although the effect of the full subsidy on voucher take-up is around 10 pp among those who are close to the health center, the size of the treatment effect on contraceptive use (4 pp) and births (2.5 pp) remains very modest.²⁸ Finally since Ashraf et al. (2014) find

²⁶Since we do not find that the voucher affected contraceptive use, we cannot estimate 2SLS models of the effects of contraceptive use on any outcome.

²⁷Among the wives surveyed at baseline, 30% agreed either with the statement that modern contraception is dangerous for health or with the statement that it causes sterility.

²⁸If we explore the heterogeneity by distance in a more systematic way, estimating treatment effects within different distance bins, we find a monotonic, negative relationship for voucher take-up, but no relationship for contraception or fertility.

that spousal disagreements matter for contraceptive take up, we also examine whether the subsidies affected the fertility of couples in which neither spouse wants another child. There is a significant effect on pregnancies, but no effect on live births or on contraceptive use, and the magnitudes of the effect remains modest. Note that for all of these subgroups, we estimate statistically significant impacts of the full subsidy on voucher use even when accounting for MHT (column 5), further confirming that the vouchers primarily subsidized those who were already using modern contraception.

In sum, we do not find any statistically significant effects of free contraception on births and modern contraceptive use even among sub-populations for which we might have anticipated larger impacts. The results do suggest there are subgroups for whom there might be modestly sized effects: those who report not being able to afford contraception and those who live less than 2 kilometers from a health center.²⁹

In the next section, we take a closer look at the idea that relaxing supply constraints is not effective when there are additional demand constraints. We investigate two specific information frictions which potentially drive up desired fertility: misperceived acceptability of birth control and misperceived child mortality.

5 The role of information frictions in explaining the lack of impact of the full subsidy

To test how the effects of the full subsidy vary with the extent of demand-side frictions, we use two approaches. First, we examine the effects of the subsidy in sub-populations that appear less constrained by the demand-side frictions we study. Second, to address the fact that baseline proxies of demand-side frictions are not exogenous, we cross-randomize information interven-

²⁹We also investigated a number of other subgroups, see [Table A.7](#). The largest impact of the full subsidy was found among those who were already using modern contraception at baseline (2.8 pp decline in births, not significant). We considered the possibility that the vouchers would be less effective in polygamous marriages in which wives' status is influenced by their fertility. We find similar null effects for both polygamous and non-polygamous couples, and for junior and senior wives. Based on the literature we also hypothesized that older women who are closer to having completed their desired fertility would be more responsive to the subsidy (in the Matlab experiment for example the point estimates suggest larger declines in fertility among older women ([Joshi and Schultz, 2013](#))). If we split the sample by median age, we find no effect on contraception and fertility in either sub-group. We also tested whether demand varies with the number of prior pregnancies but found this is not the case. The point estimates of the effect of the subsidy on pregnancy and births are small in magnitude and insignificant, even among women who already had 5 pregnancies or more. Finally, we see no increase in modern contraceptive use for this group.

tions. In this section, we start by providing summary statistics on the extent of misperceptions, then we describe the information interventions, and finally present the results.

5.1 Descriptive evidence on potential information frictions

5.1.1 Individuals' attitudes and perceived social norms

Table 4 shows that, although people know about contraception and many use it, social acceptability of contraception is perceived as low: a substantial fraction of people report that the community views its use to delay first births negatively (65%), or stigmatizes family planning users (39%). These negative social perceptions stand in contrast with private views about contraception: most individuals think modern contraception is acceptable. This mismatch could reflect either pluralistic ignorance or the fact that elders (who are not in our sampling frame) hold the more traditional views and set the community norms. Regarding family size, the vast majority of respondents agree there is a quantity-quality trade-off and do not perceive there is a social norm to have a large family.

Table 4: Survey data on potential misperceptions

	Mean	SD	N
Personal views (first-order beliefs):			
Agrees: modern contraception is not dangerous to health	0.79	0.41	13,011
Agrees: modern contraception is not against tradition	0.80	0.40	13,051
Agrees: modern contraception is a reliable way to control births	0.93	0.25	12,301
Agrees: there is a quantity-quality tradeoff	0.83	0.37	12,207
Perceived social norms (second-order beliefs):			
Agrees: times are changing and there is no social norm on family size	0.87	0.33	12,358
Agrees: community disapproves couple using contraception to delay 1st birth	0.65	0.48	14,609
Reports women sometimes punished/stigmatized for using contraception	0.39	0.49	13,353
Child mortality:			
Has had at least one child who died	0.28	0.45	14,353
True under 5 mortality rate (%)	10.52	2.09	14,609
Wife's perceived under 5 mortality rate (%)	19.81	19.47	11,271
Wife overestimates under 5 child mortality	0.55	0.50	11,271

Notes: Data from Baseline survey with wives.

Our post-intervention data provide some descriptive evidence suggesting a generational divide. Although this varies across villages, acceptability of contraception and small families is quite high overall, with more than 50% in most villages supporting them, particularly among younger individuals and women. This suggests scope for village meetings to spark cross-group dialogues, perhaps resulting in changing the norms.

5.1.2 Perceived and actual child mortality

As reported in [Table 4](#), current levels of mortality in this context remain high by international standards: around 10% of children do not survive to age 5. The rate ranges across provinces from 7% to 14%. The situation was much worse and more diverse in the past: the average was around 16%, ranging from 12% to more than 20% in several provinces, including 32% in one province.³⁰ Almost 30% of women in our sample have had a child who died, so child mortality is a salient phenomenon.³¹ The beginning of the twenty-first century was thus a period of progress and convergence. Current levels and trends are consistent with those observed for the country as a whole in the Demographic and Health Surveys.

How do people form expectations in this rapidly evolving context? We piloted different ways of eliciting subjective expectations and opted for a straightforward question: *According to you, out of 100 children born today in your village, how many will survive to the age of 5?* Respondents overestimate current mortality levels in all provinces but one: the average perceived risk is 20% (see [Figure A.7](#)). This is consistent with demographers' hypothesis that there is a lag between perceived risk and actual risk because it takes time to adjust to mortality changes ([Montgomery, 2000](#)). An alternative explanation is that people overestimate small probabilities, and since mortality is declining, the perceived risk is mechanically closer to mortality rates in the past than today. Our evidence that on average there is over estimation is consistent with studies by [Delavande and Kohler \(2009\)](#) in rural Malawi, and [LeGrand et al. \(2003\)](#) in Zimbabwe and Senegal who also find that people tend to overestimate child mortality on average.

Given heaping in responses, we classify responses that are within 5 percentage points (pp) of

³⁰[Figure A.7](#) plots the under-5 mortality rates estimated from our listing data in different provinces for the most recent cohorts (born between 2007 and 2012) and older cohorts (born between 1973 and 1998). This is precisely the information provided in the information treatment arms.

³¹The 30% rate is consistent with most women having had three children at baseline. With a 10% child mortality rate, the odds that at least one of them has died is 27% (assuming independence).

the true rate as accurate. About a third of beliefs are accurate (see [Figure A.8](#), which plots the distribution of perceived mortality rates by gender). About half of respondents overestimate the mortality rate by more than 5 pp. Many of them make large mistakes: the mean perceived rate is 19% for women and 16% for men. The remainder of people underestimate the actual rate by more than 5 percentage points.

To recap, while some people greatly overestimate child mortality, overestimation is far from universal. This has implications for the expected sign of the information treatment’s effects. The information can help everyone re-calibrate their beliefs, with heterogeneous effects on subsequent behavior, depending on whether people underestimate or overestimate at baseline.

5.1.3 Heterogeneity analyses

We assess whether the demand for contraception and fertility effects of the full subsidy are larger among women who at baseline either: (a) do not believe that their community disapproves of contraception, or (b) do not report being punished or feeling stigmatized for using contraception. The results in [Table 5](#) show that there are no statistically significant effects of the free contraception among these groups (all of Anderson’s q-values in columns 1 to 4 exceed 0.10), and the magnitude of the fertility effects is modest (a decrease by 2.7 pp (=4.3%)).³²

Next, we assess the role of perceived child mortality rates. Perhaps women do not want contraception because they perceive child mortality as higher than it is and, thus, desire a large number of children. In this case, free vouchers would be ineffective. However, [Table 5](#) shows that the full subsidy had no impact on births (a magnitude of 1.6 pp or 2.5%) in the group of women who do not over-estimate child mortality. This would suggest that factors other than child mortality keep the demand for children high and/or the demand for modern contraception low.

However, these results are difficult to interpret because they are based on baseline characteristics that are not randomly assigned. To overcome this limitation, we now make use of the other randomized interventions we implemented which aimed to lower demand-side barriers.

³²See [Table A.8](#) for the specification with baseline controls.

Table 5: Treatment effects when potential misperceptions are absent

		(1)	(2)	(3)	(4)	(5)
Sub-sample		Had a live birth since baseline	Had a pregnancy since baseline	Used medical contraception in last 3 yrs	Month(s) used modern contraception (last spell of each type) in last 3 yrs	Used IPA subsidy voucher
Social Norms:						
Does not believe community disapproves use of contraception (N=4,277)	Full Subsidy	-0.027*	-0.029**	-0.005	-0.534	0.043***
		(0.014)	(0.014)	(0.019)	(0.509)	(0.013)
	q-value	0.101	0.101	0.789	0.440	0.005
	Control Mean	0.627	0.703	0.531	9.600	0.150
Does not report women being punished or stigmatized for using contraception (N=6,855)	Full Subsidy	-0.027*	-0.027*	-0.013	-0.609	0.030**
		(0.015)	(0.015)	(0.016)	(0.464)	(0.012)
	q-value	0.101	0.101	0.643	0.440	0.012
	Control Mean	0.634	0.714	0.559	10.239	0.155
Mortality Perceptions:						
Does not overestimate under-5 child mortality (N=4,227)	Full Subsidy	-0.016	-0.005	0.031	0.411	0.034***
		(0.016)	(0.017)	(0.019)	(0.529)	(0.012)
	q-value	0.331	0.780	0.307	0.440	0.011
	Control Mean	0.628	0.707	0.527	9.473	0.148

Notes: Endline controls are: Whether the village had to be surveyed by phone due to security concerns at endline and date of the endline survey. Province fixed effects are used in all specifications. Robust standard errors in parentheses. Clustering at the health center level. Control Mean refers to that of 10% subsidy sub-sample. N refers to the number of observations in each subsample. The number of observations may vary slightly between outcomes; in this case, we report the minimum number across all 4 regressions. q-value refers to the adjustment for Multiple Hypothesis Testing in [Anderson \(2008\)](#). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

5.2 Interventions to address potential frictions

5.2.1 Village meetings

The goal of the village meetings was to expose participants to both sides of the debates regarding contraception and fertility, generate community dialogue, and create common knowledge about community views. We implemented two types of meetings, where people had either an active or a passive role.

Village debates. Villagers were invited to a public meeting and asked to volunteer for a debate. Six teams (with 5-7 members, diverse in age and gender composition) were formed and assigned a specific position, creating three debates on (1) the quantity-quality trade-off, (2) the relevance of pro-natalist norms in a changing world, and (3) the pros and cons of modern contraception.³³ Teams prepared their arguments for about 15 minutes and then debated in

³³Debates were framed as follows. **1:** Having plenty of children makes you feel rich and proud. The more children you have, the more likely you are that one of them will succeed *versus* It is not quantity of children that matters, but whether they surpass you. If you have many children, it is difficult to give everyone a good chance in life. **2:** Having a lot of children gives prestige. Even couples who would prefer a small family should have as many kids as everyone else; otherwise they will get ostracized *versus* Each couple is in the best position to know what is best for their individual case. The community appreciates everyone. **3:** If a couple wants to wait several years before the next birth, they should use traditional means of birth control. Modern

public for about 15 minutes. After both teams had presented their case for each of the three debate prompts, all attendees were asked to cast an anonymous vote for the position they personally agreed with. Men and women of older and younger generations cast their votes in separate urns, and facilitators tallied votes and announced final vote shares by gender and age group. Note that unless all individuals within a group voted identically, no one’s private vote could be inferred.

Village edutainment. Villagers were invited to a public meeting where a 30-minute film was shown. The film depicted villagers engaging in the same three debates on quantity-quality trade-off, pro-natalist norms, and modern contraception, in a more naturalistic, conversational way. Again, at the end of the film, meeting attendees cast anonymous votes and tallies were publicly reported. To write the script of the movie, we piloted similar debates in other villages in 2017 and used the arguments often mentioned by participants. We then hired a professional filmmaker and actors to create the film. For the screenings, we partnered with Cinema Numérique Ambulant, an NGO dedicated to bringing cinema to remote areas throughout West Africa.³⁴ An advantage of the film over the debates was that it ensured there were no false but purportedly factual statements made.

The village meetings could influence fertility through several mechanisms. First, seeing both sides of a controversial topic may prompt participants to change their views and behavior, independent of social interactions. Second, the discussion made these topics salient and could prompt further discussion among people to persuade one another. Third, the votes generated common knowledge about the distribution of views in the community, solving potential information failures regarding others’ views. In this context, women rarely publicly speak up in the presence of men, as is also the case for younger people in the presence of older people. Thus, revealing votes by gender and age might reveal heterogeneity across groups who usually have limited conversation on these topics. Fourth, the common knowledge and discussion “from the bottom up” with the participation of village leaders could enable people to coordinate, for example with younger people deciding to disregard views mainly held by elders.

Individual edutainment. Some but not all of the hypothesized mechanisms through which the village meetings could change behavior are collective in nature. To help separate

contraception is a danger to the family *versus* They should use modern contraception; this is the most reliable way to control and space births.

³⁴The film can be seen at this [link](#).

individual-level belief change from social/collective mechanisms, we followed [Bidwell et al. \(2020\)](#) and implemented an individual treatment in which couples (including co-wives in polygamous households) saw the edutainment film at home.

Public meetings were successfully implemented in 297 out of 300 villages in the spring 2018.³⁵ Sampled households were informed at the end of the survey about the village meeting, and local leaders were asked to diffuse the information to the rest of the village. Attendance at the start of the meetings averaged 95 people [median: 75, range: 23 to 299], representing roughly 25% of the village population, and was almost identical between debates and edutainment villages. People typically stayed until the end, and additional people often came as the event progressed. Afterwards, many villagers expressed their satisfaction to the implementing team.

5.2.2 Mortality information

This intervention provided information about the province-level child mortality rate and its recent decline. We chose to provide province-level information rather than national-level information because there is a lot of geographic heterogeneity in mortality rates within Burkina Faso, and we thought the risk of providing locally-irrelevant information was high with national statistics alone.

The statistics were computed using birth history modules (similar to those used in the Demographic and Health Surveys) collected in sampled villages during the listing phase. We collected the data ourselves because province-level data on child mortality was unavailable with enough precision in any existing dataset. One attractive feature of the process we used to generate the information is that it was done in partnership with the very population with whom we were working. People may be more likely to comprehend and trust information when they are involved in the information generation process.

The mortality information was delivered by a trained enumerator at the end of the baseline survey. All adults in the household who were present were invited to listen to the information and were given the chance to ask questions. The script also described the source of the data: *“Recall we came here a few months ago to ask some women in your village about their birth history. We have compiled this information and can now report on the results”*. To maximize

³⁵Three villages refused to host a meeting before knowing which type of activity would be proposed. They were all in the “Debates” arm. These villages are, of course, kept in the analysis (the results sections below present intent-to-treat estimates).

comprehension, the information was presented verbally and visually. [Figure A.9](#) displays an example of the charts shown by enumerators and explained as follows: *“The picture on the left shows the situation in the past, and the picture on the right shows the situation today. All the children in purple died before age 5, while all the children in yellow survived until age 5. As you can see, for the previous generation of mothers, such as your mother, 21 children out of 100 died. For the current generation of mothers like yourself, only 11 died. This means that there are 10 children who would have died in the past but, because of the recent progress, they were able to survive.”* Enumerators asked a series of follow-up questions to gauge the reaction to the information. Over 95% reported understanding the information. About 50% described the information as “new”.

5.3 The effects of contraception subsidies when potential information frictions are addressed

We estimate a version of equation 1 where we interact the subsidy with dummies for different demand-side treatment arms. The results are presented in [Table 6](#). Results for the specification including baseline controls are similar and reported in [Table A.9](#). We pool the debates and edutainment in the analysis because they have a mostly common theory of change, and we see no difference between them in their effects (see [Table B.2](#)).

5.3.1 Providing information about opposing views and social norms

By testing if the combined interventions (village meetings and free contraception) reduce fertility, we are testing the joint hypothesis that (a) misinformation about attitudes and norms systematically increases desired fertility and reduces acceptance of modern contraception under the status quo and (b) cost is a barrier to contraceptive use.

We do not find that the village-level interventions modified the effects of the full subsidy: none of the interaction terms are significant ([Table 6](#)). The main effects of the interventions themselves are also insignificant, suggesting that the interventions had no average effect. We confirm this when we estimate the effects of the demand interventions without interaction terms ([Table B.3](#)). We find similar null results for the individual edutainment treatment, which entailed showing the woman and her household members the edutainment film on a tablet computer in her home. Importantly, the interventions did not change either individual beliefs

or perceptions of social norms, either on average or based on initial beliefs (Table A.10).³⁶

The absence of an average effect could mask some people updating and changing their behavior in either direction. However, when we examine heterogeneity by baseline personal views or by beliefs about social norms, we also see no effect (Table A.11). The lack of effect cannot be attributed to poor implementation: Attendance at the meetings was high, so people did hear others' views but did not change their own views as a result.

5.3.2 Addressing mortality misperceptions

In the same way, the full subsidy did not lead to a larger decline in fertility for those provided accurate child mortality information. The coefficient on the interaction is small and statistically insignificant for births and pregnancies (Table 6 columns 1 and 2). The interaction term is also small and insignificant for the contraceptive use measures (columns 3 and 4). Table 6 further shows that the intervention itself did not affect fertility or the demand for contraception (the main effects are not significant).

This null effect is not due to some people revising their mortality beliefs downward while others revised upward. Figure A.10 shows the distribution of perceived minus actual child mortality at endline among women, for the treatment and comparison groups. If people updated toward the statistics provided, we would expect the treatment group to have a narrower distribution, bunched closer to a zero gap between actual and perceived mortality. We see no such narrowing. Furthermore we cannot reject that the distributions are identical.³⁷ Thus, this intervention did not work in the sense that the first step in the theory of change—updating beliefs—did not occur.

³⁶One of the twenty reported treatment and treatment interaction coefficients is significant at the 5% level.

³⁷The average under-5 mortality is perceived to be 16.4 percent at endline. The individual information treatment lowered this by 0.29, or less than 2%. The null effect holds for men as well. When we split the sample by whether individuals initially over- or under-estimated child mortality, we find no effect on perceptions for either group. The effects of the mortality information treatments are also small and insignificant among subgroups with the least reason to disregard the information we provided, such as those who had never lost a child (Table B.4). Finally, when we compare the effect of the village debate and the debate with mortality information, there is no difference in beliefs.

Table 6: Treatment effects on primary outcomes: fully interacted

	(1)	(2)	(3)	(4)	(5)
	Had a live birth since baseline	Had a pregnancy since baseline	Used medical contraception in last 3 yrs	Month(s) used modern contraception (last spell of each type) in last 3 years	Used IPA subsidy voucher
Full Subsidy	-0.036* (0.019)	-0.025 (0.018)	0.006 (0.023)	-0.280 (0.542)	0.047*** (0.017)
Village Debate or Edutainment	-0.010 (0.017)	0.008 (0.015)	0.012 (0.019)	0.291 (0.484)	0.026** (0.012)
Individual Edutainment	0.006 (0.022)	0.003 (0.020)	-0.006 (0.021)	0.393 (0.627)	0.013 (0.015)
Individual Mortality Info	0.010 (0.025)	0.008 (0.021)	0.025 (0.022)	0.886 (0.665)	-0.015 (0.014)
Village Interventions X Full Subsidy	0.023 (0.023)	0.001 (0.022)	-0.011 (0.029)	0.117 (0.691)	-0.027 (0.021)
Individual Edutainment X Full Subsidy	0.031 (0.031)	0.044 (0.030)	0.030 (0.031)	-0.285 (0.831)	-0.024 (0.025)
Individual Mortality Info X Full Subsidy	0.027 (0.034)	0.009 (0.031)	-0.025 (0.033)	-0.728 (0.842)	0.017 (0.022)
Province FE	Yes	Yes	Yes	Yes	Yes
Baseline Controls	No	No	No	No	No
Observations	12,542	12,543	12,131	12,107	12,519
Control Mean	0.628	0.701	0.526	9.373	0.129

Notes: Endline controls are: Whether the village had to be surveyed by phone due to security concerns at endline and date of the endline survey. Province fixed effects are used in all specifications. Robust standard errors in parentheses. Clustering at the village level. Control mean refers to that of the pure control. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Why did individuals not update their beliefs in response to the information? Study participants seem to have understood the information, so the explanation seems to lie elsewhere. One possibility is that they did not believe the information, perhaps because it conflicted with their experience. Recall that 30% of women have lost a child. The first-born mortality rate is higher than the mortality rate for later-births, which mechanically makes the rate experienced so far by a cohort higher than what they should expect going forward. Another possibility is that the political instability and COVID-19 pandemic made people fearful that the recent progress would be erased, rendering the information on the recent past less pertinent. While we can only speculate why people did not update, our results highlight that it is very difficult to change beliefs about child mortality.

5.3.3 The effect of subsidies in the absence of demand interventions

The coefficients on “Full subsidy” in Table 6 indicate the effect of the subsidies in the absence of any demand intervention, comparing the pure control households and the households who

received only the subsidy, as shown in [Figure 1](#). In column 1 (live birth), the coefficient is significant at 10% and larger than in the non-interacted specification reported in [Table 2](#) (3.6 pp against 1.7 pp). This result could be interpreted as suggesting that, in the absence of demand-side interventions, the subsidy would have resulted in a decrease in births, and that the demand-side interventions played against the subsidies. However, the estimated impact is still modest: the point estimate translates into a 5.7% decrease in births.

In addition, the results in columns 2, 3 and 4 show that there is no statistically significant effect on pregnancy and modern contraceptive use. In magnitude, the coefficients are very similar to what we find in [Table 2](#). Moreover, we do not find evidence that there were effects of the demand-side interventions on beliefs or outcomes, so it is not clear why we would want to concentrate attention on the subgroup that did not receive them. That is why our preferred estimate remains the coefficient in [Table 2](#), which has the greatest statistical power.

5.4 Discussion: The residual explanation for high fertility may be high desired fertility

Our finding that free modern contraception did not influence fertility, even among subgroups with ex-ante high demand for contraception (based on the conventional measure of unmet need) or after addressing potential frictions influencing desired fertility, suggest that it may be time for policy-makers concerned about the “stalled” fertility transition to start suspecting the obvious: households *want* many children. This is, after all, what they say when asked about their ideal family size: the mean is 9 for men and 6 for women in our data, and similar levels are found in DHS surveys.

Our survey data lends support to the view that there are still high returns to having many children in the context we study. First, individuals cite their need for old age support. [Table A.12](#) shows that 74% of men and 81% of women report they would not be able to cope in old age without the support of children. Moreover, not all children are able to support their parent, and a common view is that each child is a “lottery ticket”: the more children one has, the higher the chances that at least one of them succeeds and can provide old age support. Men expect only 20% of their children to send enough money back to sustain parents, for example. Second, children are also an important source of labor even before their parents reach old age. 39% of household heads report that family labor is the main constraint to expanding their farm

size. When asked what would happen if they had one fewer child than they desire, 17% of women say that they would not have enough children to perform tasks around the house and 13% would not have enough children to work in their agricultural fields. Finally, children are a source of happiness. 14% of women say that they would be less happy if they had one less child than they want. Altogether this qualitative evidence suggests that households want many children and that the value of additional children may exceed the value of investing more in each individual child in rural Burkina Faso.

6 Conclusion

This study investigated the effectiveness of providing free contraception for three years in rural Burkina Faso, a context where both desired fertility and realized fertility are among the highest in the world. We found a precise null effect of offering free contraception on birth rates. This is an important finding from a policy perspective given that subsidized access to modern contraceptives is a major focus of women’s empowerment programs around the world, among governments and international organizations alike. Our results suggest that, in contexts similar to rural Burkina Faso, the relationship between free family planning, women’s empowerment and economic development is less straightforward than usually assumed.

Our results do *not* imply that efforts to improve access to modern contraception have no value. First, reproductive freedom is an important right. Second, some women, even if not many, may be financially constrained. Free access to contraception may enable them to reduce their fertility to their desired level or to better manage the timing of fertility. Third, we studied only married women. Access to contraception might affect age at marriage and delay first births, even without affecting the total fertility rate. It is also possible that free family planning will change attitudes among people who grow up in this regime, leading to larger long-run effects. Fourth, we studied the complementarity between subsidies and two specific information treatments. Complementarities may exist with interventions targeting other non-financial barriers identified in the literature, such as distance/convenience, misperceptions of side effects and female mortality risks, or intra-household bargaining.

Our study provides other lessons. First, even when people overestimate child mortality, providing accurate, relevant and reliable information on levels and trends did not influence mortality beliefs in the medium run, and did not increase the take-up of modern contraception

even when it was offered for free. Research remains to be done to better understand how survival expectations are formed and most importantly how they may be modified – our results suggest that this is not an easy or cheap task since our efforts to collect and disseminate information locally failed. Second, while social norms and perceptions of these norms are important, there do not appear to be large misperceptions regarding these norms in our context. As a result, the provision of free modern contraception was not more effective in communities with debates or movies addressing social norms. One-time community-level interventions may not be a useful policy lever. Future research should further investigate the conditions under which price reductions can be effective.

Do these results, especially the lack of responsiveness of fertility to contraception access, suggest a “West Africa” exceptionalism? Our reading of the evidence from other contexts does not suggest so. In higher-income countries the introduction of the pill is thought to have had modest effects on aggregate declines in fertility ([Bailey, 2012](#); [Knowles Myers, 2017](#)), and these effects, moreover, were found in contexts where legal and social change was occurring in tandem ([Goldin and Katz, 2002](#)). In lower-income countries, family planning programs also had a modest impact on fertility decline, with some exceptions ([Miller and Babiarz, 2016](#)). Reflecting on the historical experience in the 1960s, [Hartman \(1997\)](#) concludes that international efforts (such as those by USAID) to expand access to contraception in low-income countries played a limited role in lowering fertility and curbing population growth. Conversely, in high-income countries, policies that provide financial incentives to households to have more children have small effects (e.g. [Kim 2020](#), [Aizer et al. 2020](#)). One interpretation of this evidence is that financial levers can only affect fertility in populations that desire to change their fertility but are prevented from doing so by financial constraints. This group appears to be small in our context, which is similar to the context in most West African rural areas. We conclude, contrary to the conventional wisdom in policy circles but in line with early contributions by [Becker \(1991\)](#), [Easterlin \(1975\)](#) and [Pritchett \(1994\)](#) and recent empirical evidence by [Dupas et al. \(2024\)](#), that fertility levels are primarily determined by deep economic factors.

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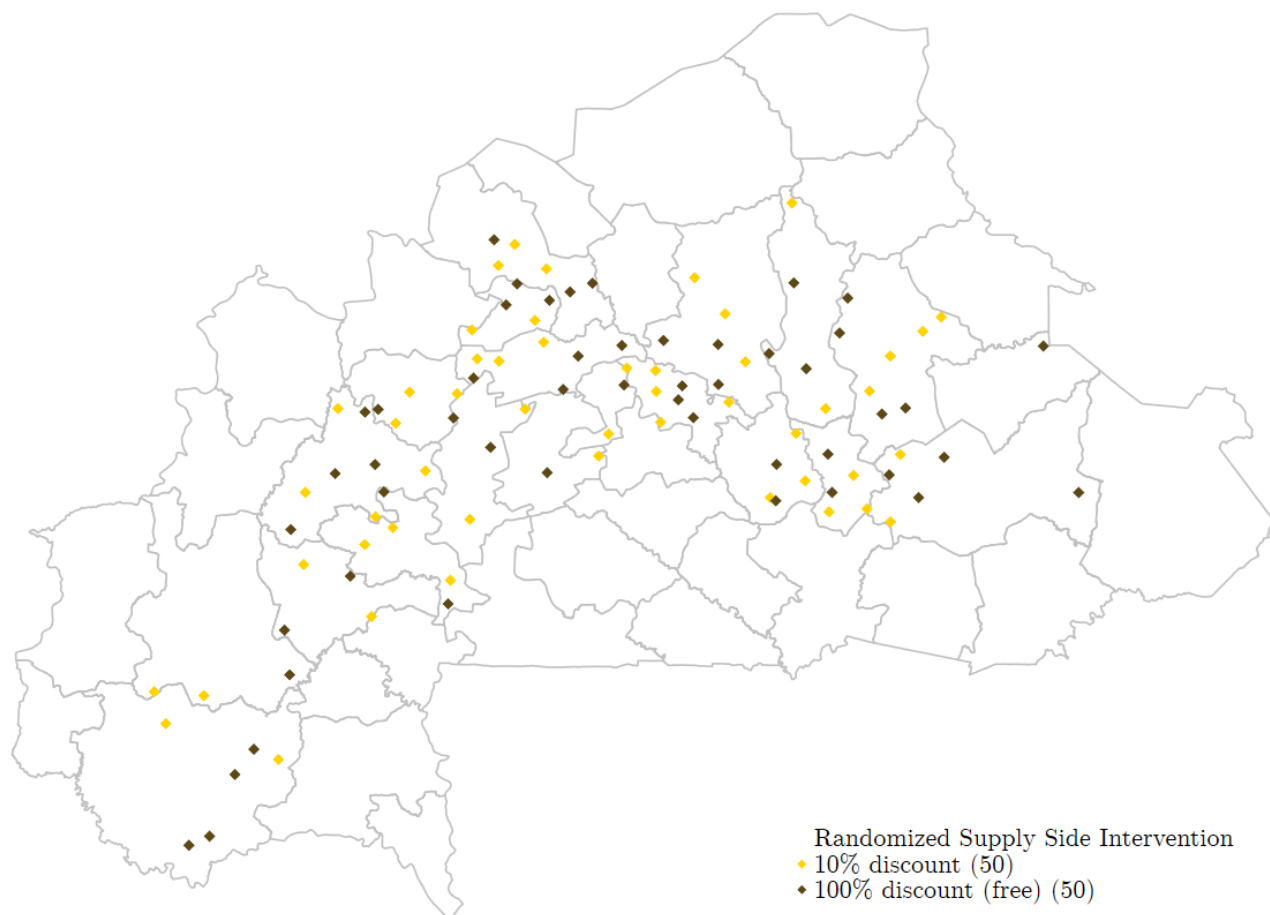
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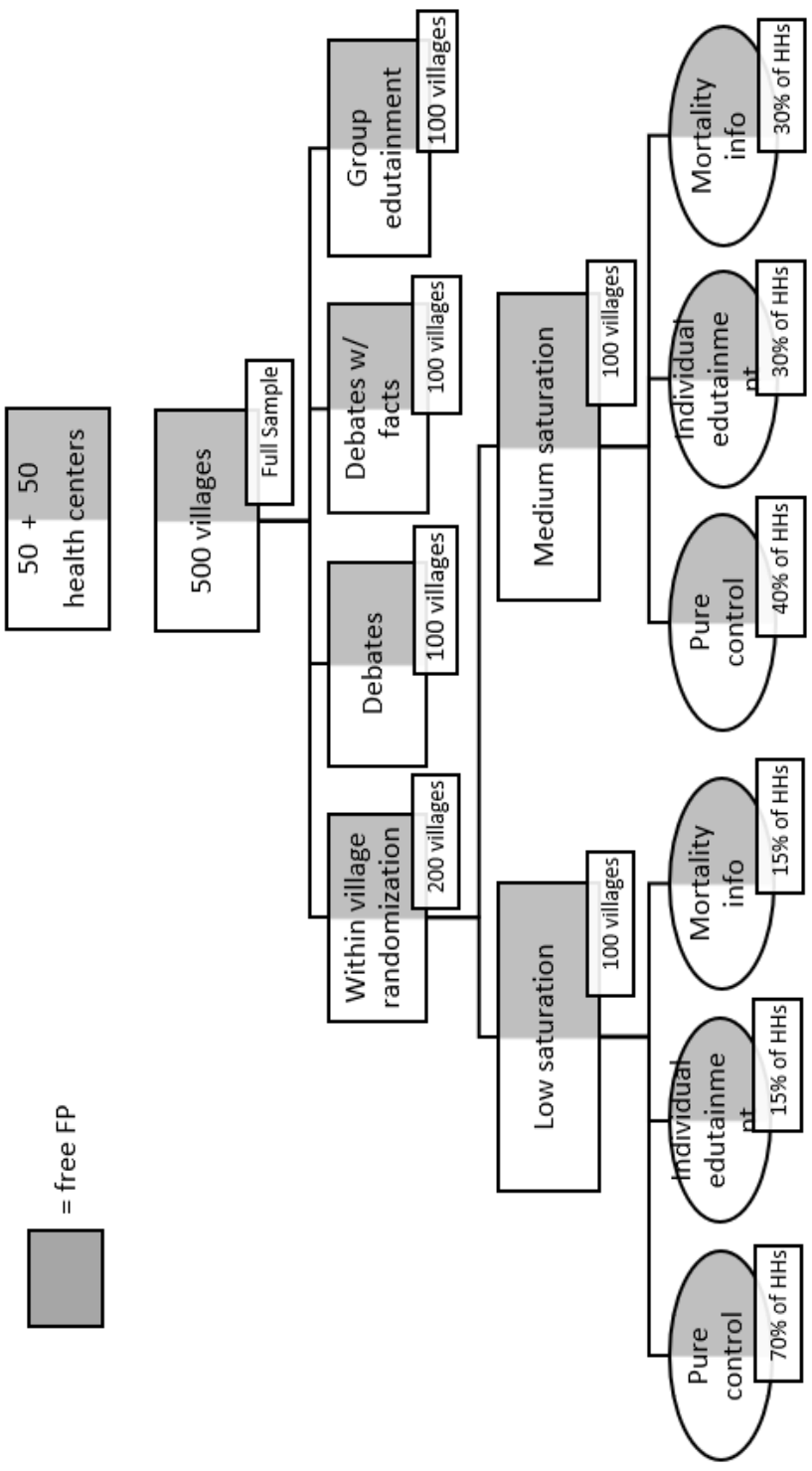
Appendix A

Figure A.1: Geographical distribution of sample and randomized subsidy treatment



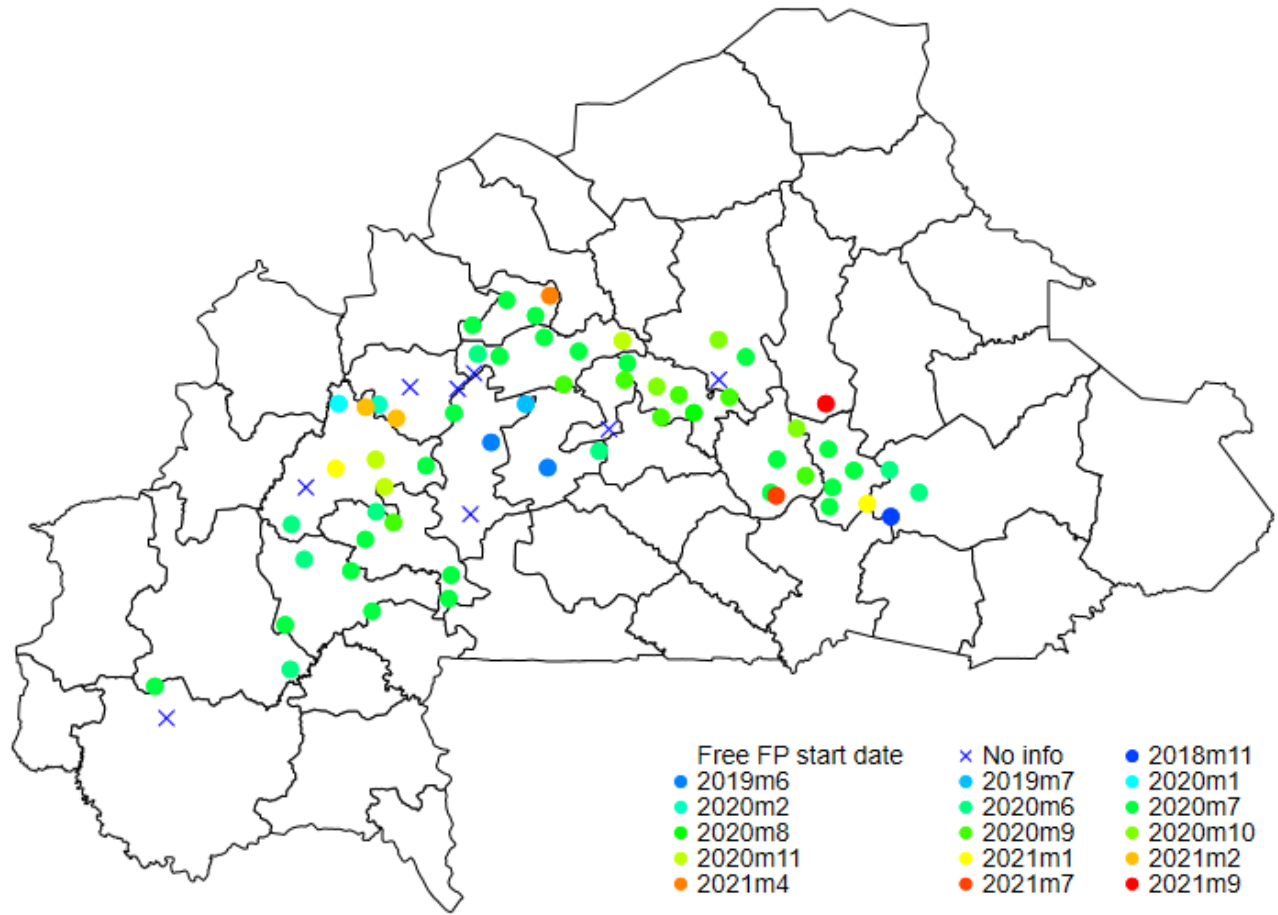
Note: Each dot represents a health center. N=100. The inner borders drawn correspond to the 45 administrative provinces.

Figure A.2: Detailed experimental design



Notes: The figure presents a more detailed version of the experimental design compared to Figure 1. There are two additional elements. (1) We vary the share of households treated within the village to assess the extent of spillovers. The medium saturation arm (village 1) assigns 30% of households to mortality info, 30% of households to edutainment, and 40% to be pure controls. In the low saturation arm (village 2), these rates are 15%, 15% and 70%, respectively. We exclude the analysis of spillovers from this paper. We find no evidence of spillovers. (2) For each center, we assign village 3 to group edutainment and villages 4 and 5 to debates. A concern about the unscripted, participatory nature of debates is that participants might make factually incorrect statements. Thus, in half of the villages with debates (village 5, debates with facts), the project staff member shared factually correct information about child mortality levels and trends, which is a topic where there was scope for participants to make untrue claims. The mortality information was delivered after the debates and facilitators used the same script and charts (printed on large posters) as in the individual information treatment. We find no difference in the effects of the debate arm, the individual mortality information arm and the debate + mortality information arm (see breakdown of the results in Table B.3).

Figure A.3: Start dates of Free Family Planning Program as reported by CSPS



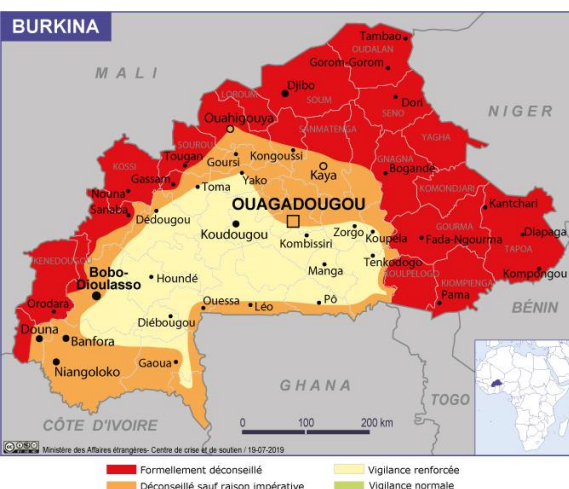
Note: The inner borders drawn correspond to the 45 administrative provinces. Officially, the program was piloted in July 2019 in two regions (Cascades and Centre Ouest) and scaled up in July 2020. The map reports the actual date when the program started being implemented in health care centers included in the sample. The data was collected through in-person visits conducted at endline. The information is missing for facilities located in areas surveyed by phone; they are not shown on the map. The baseline and interventions took place between February and June 2018 and the endline took place between February and June 2021. See [Table A.7](#) (second-last row) for the results excluding the regions where the national free family planning program was piloted in 2019.

Figure A.4: Gradual worsening of security situation in Burkina Faso over the study period

(a) June 2017



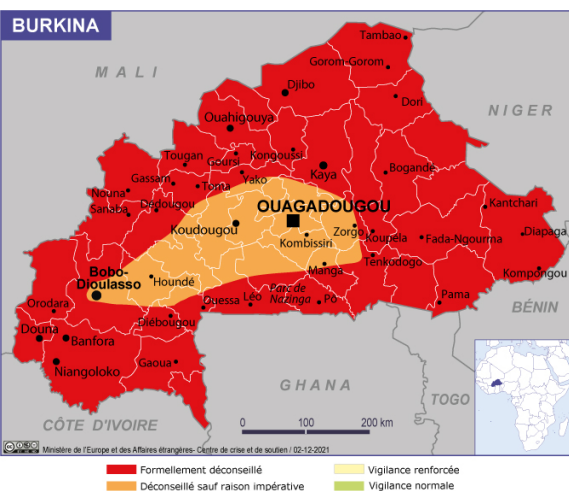
(b) July 2019



(c) December 2020

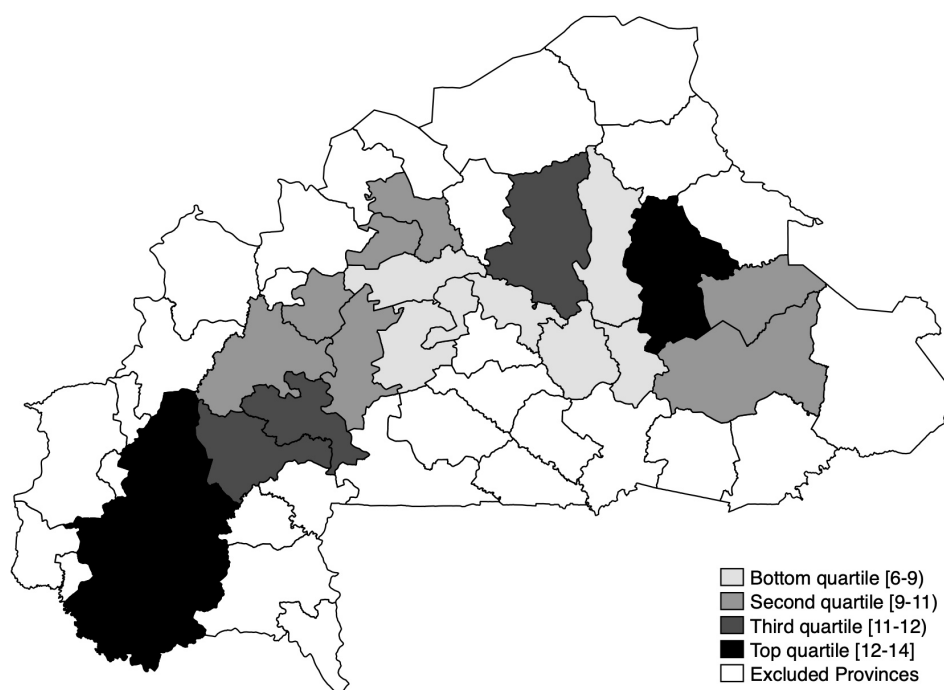


(d) December 2021

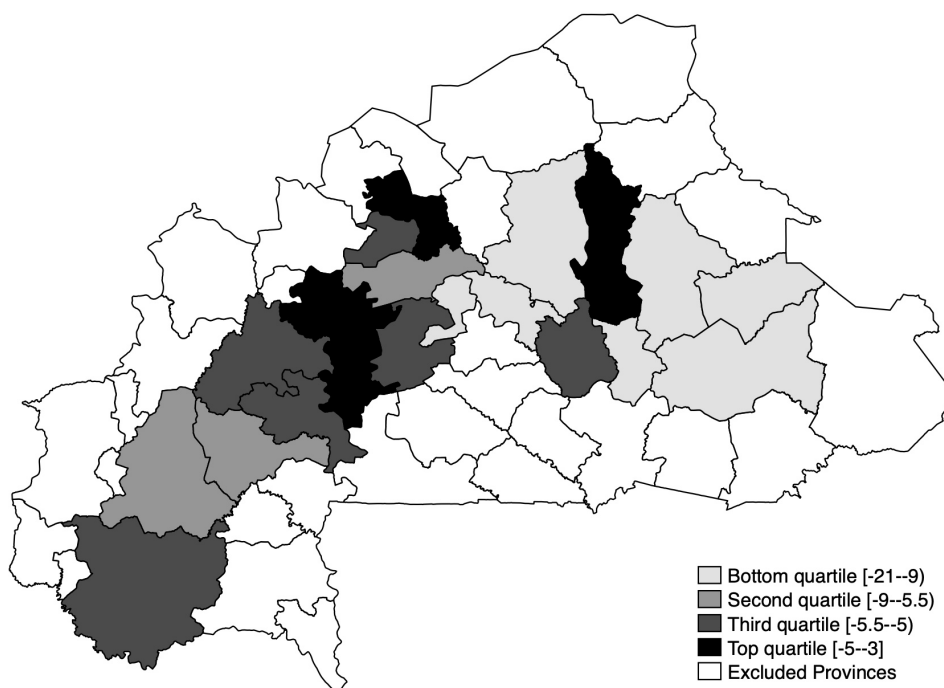


Notes: Maps published by the French embassy in Burkina Faso. Red corresponds to areas where the recommendation is “No travel”. Orange corresponds to areas where travel can be considered only under special circumstances. Source: <https://www.diplomatie.gouv.fr/fr/conseils-aux-voyageurs/conseils-par-pays-destination/burkina-faso/#securite>

Figure A.5: Under 5 mortality rates by province: levels and trends



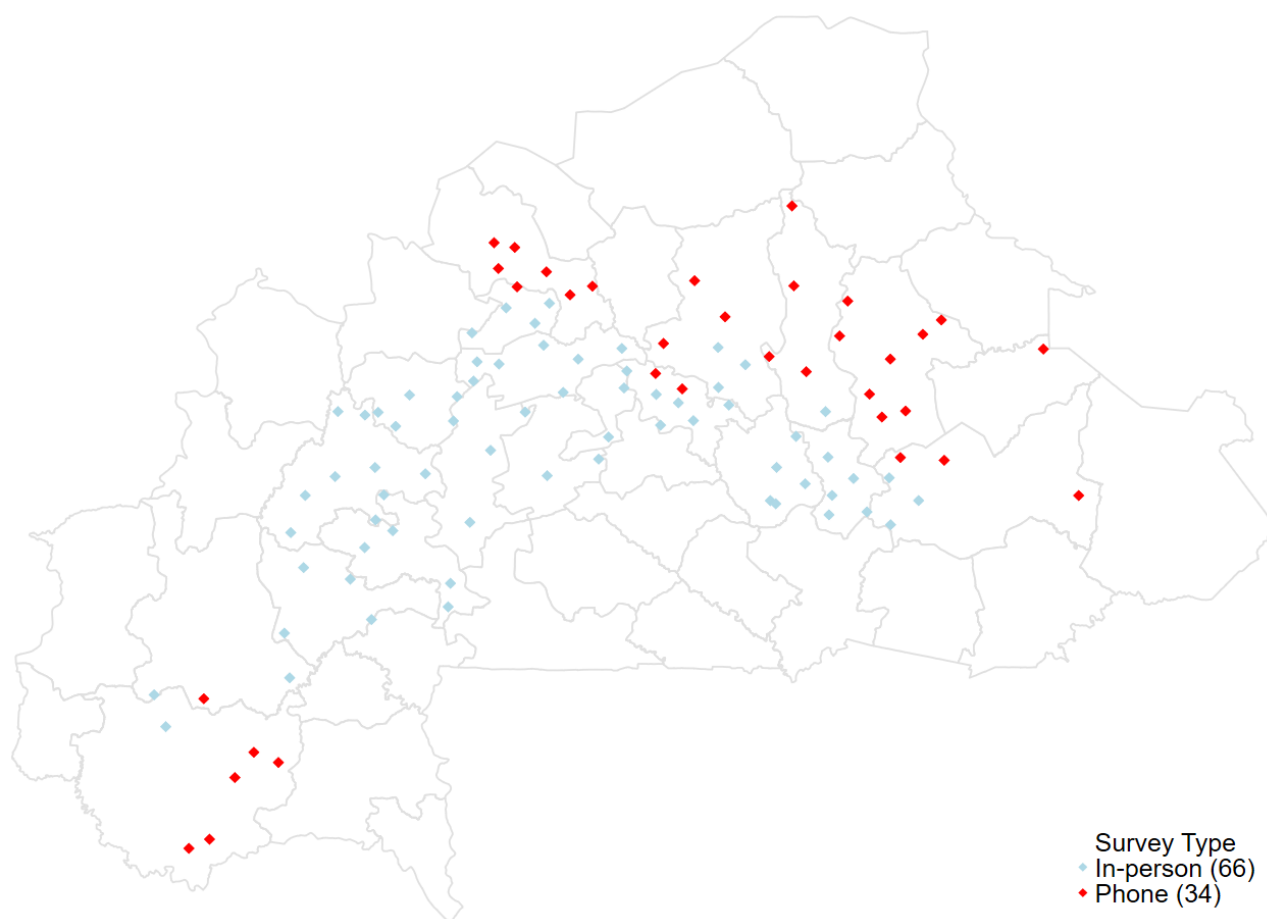
(a) Recent levels



(b) Recent trends

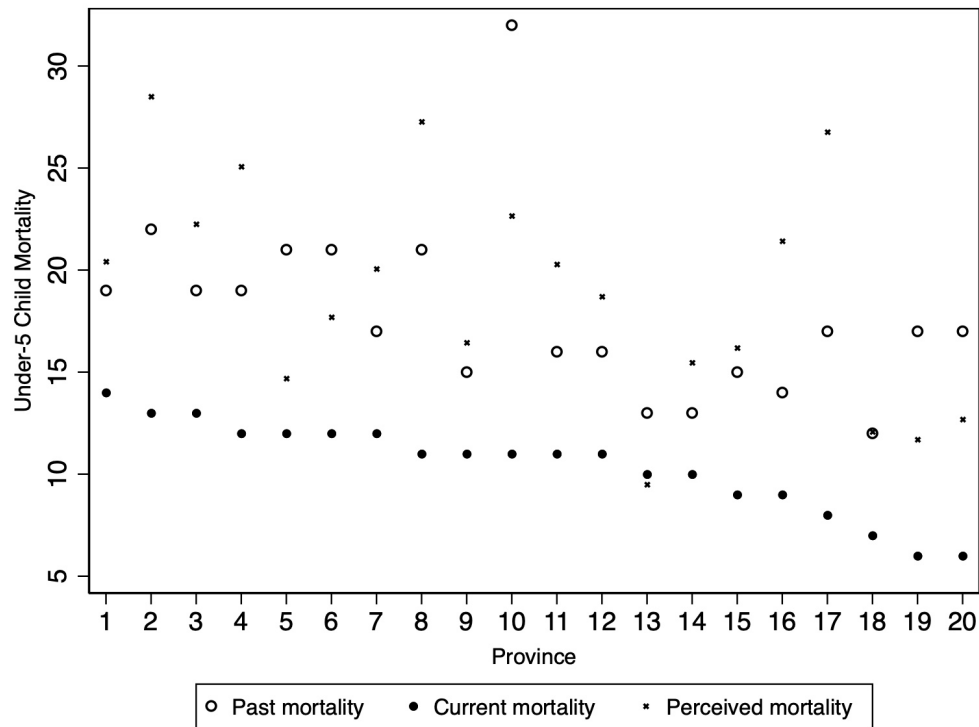
Notes: Source: Listing data. Graph (a) shows the estimated under 5 mortality rates for cohorts born between 2007 and 2012, by province. Darker colors indicate higher rates. Graph (b) shows the estimated change in under 5 mortality rates between cohorts born between 2007 and 2012 and cohorts born between 1973 and 1998, by province. Darker colors indicate smaller declines.

Figure A.6: Location of in-person vs. phone survey at endline



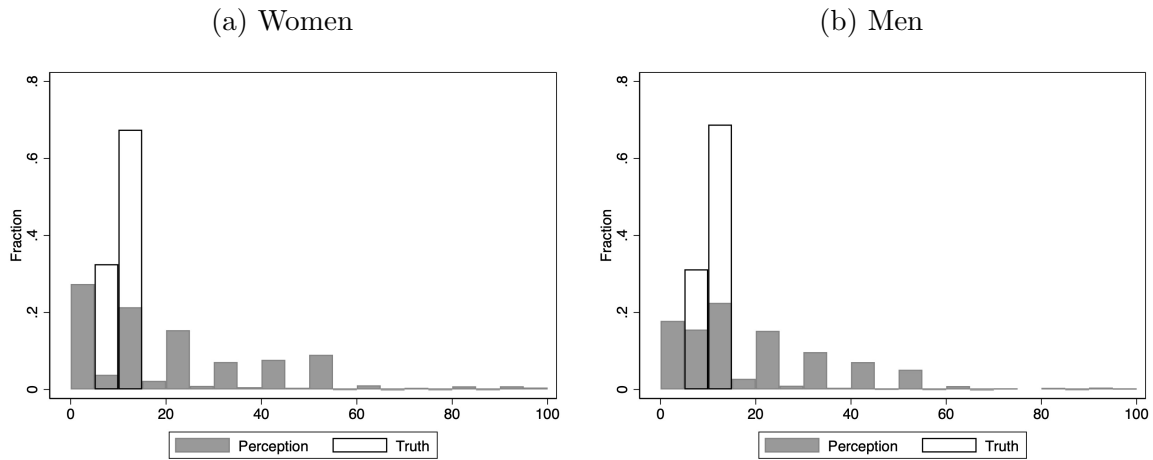
Note: Each dot represents a health center. N=100. The inner borders drawn correspond to the 45 administrative provinces. All respondents in all villages assigned to a given health center were surveyed using the same mode. The decision of which center could not be surveyed in person due to security concerns was made by the Burkina Faso office of Innovations for Poverty Action.

Figure A.7: Baseline mis-perceptions of the child mortality rate



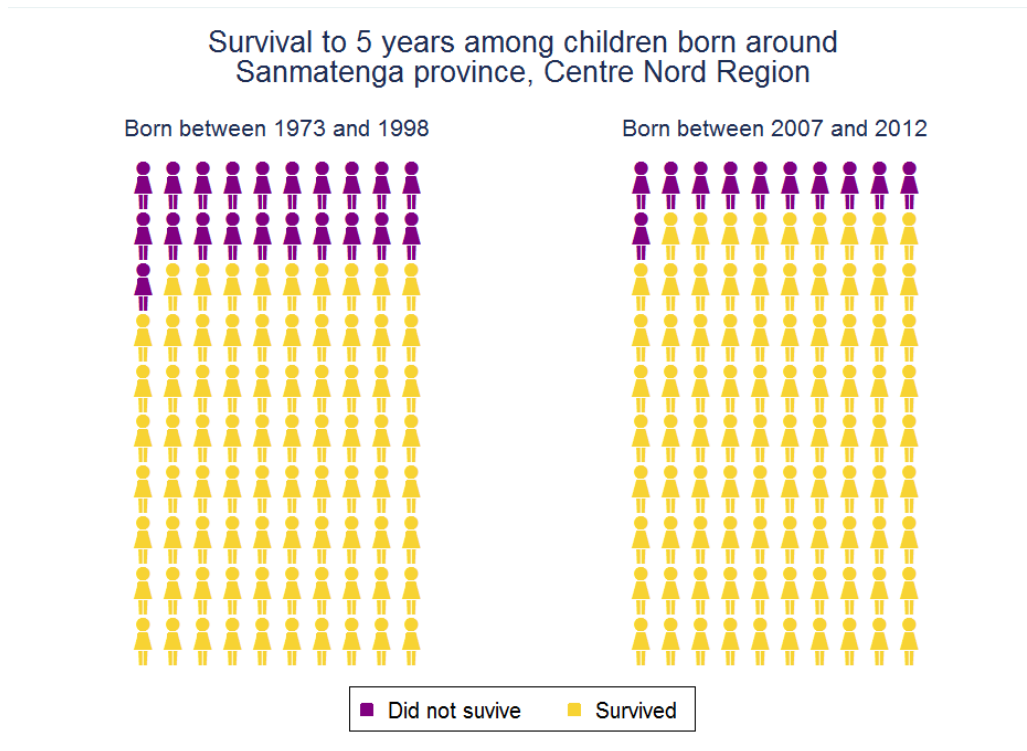
Notes: We compare the average perceived rates reported by women surveyed at baseline (shown with crosses, N=11,298) with the observed rates measured during the listing, by province. The solid dots represent the most recent cohorts, born between 2007 and 2012. The hollow dots represent older cohorts, born between 1973 and 1998. Provinces are ordered by recent mortality rates.

Figure A.8: Distribution of perceived vs actual mortality at baseline



Notes: The white bars show the distribution of the actual mortality rate measured during the listing for children born in 2007-2012 (average= 10.5; median= 10). The grey bars show the baseline distribution of the perceived rate reported by women in graph (a) and by men in graph (b). The bins are as follows: [0,4], [5,9], [10,14] ... [90,94] and [95,100]. Rates are expressed in percentage points.

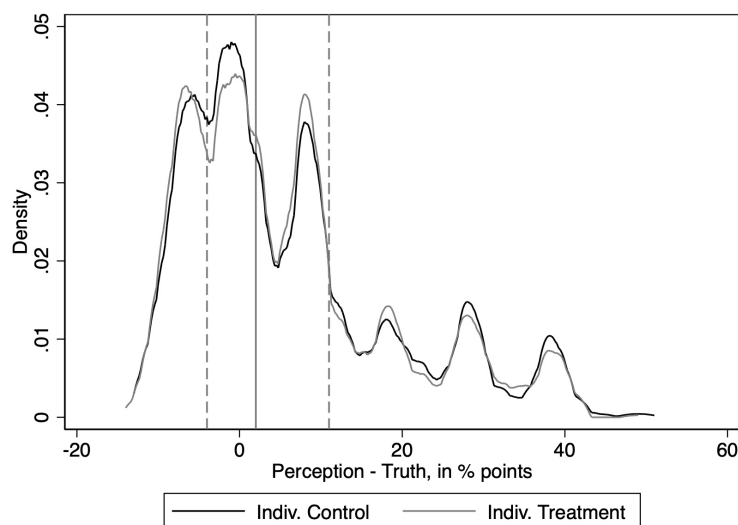
Figure A.9: Area-specific infographic used for mortality information treatment



Notes: We created one such graphic for each of 20 provinces based on data from over 190,000 births collected during the listing survey. See section 5.2.2 for details.

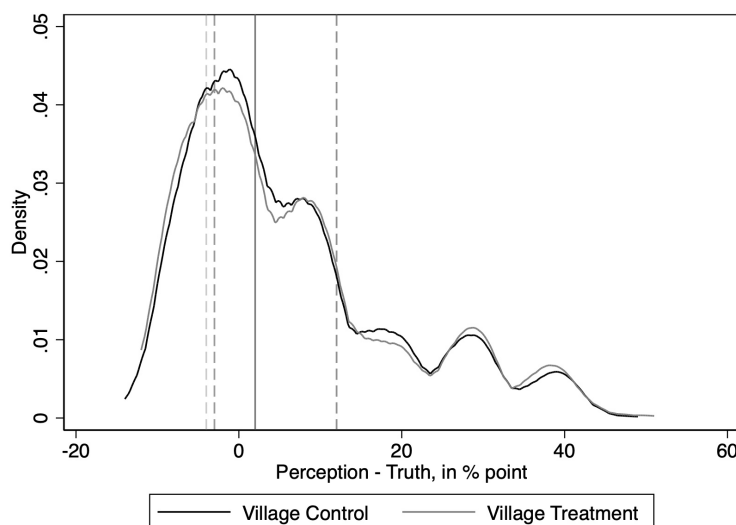
Figure A.10: Distribution of the gap between perceived and actual mortality at endline

(a) Individual Mortality Info Treatment versus Control



Note: Treatment (N: 1290, average=5.9), Control: (N : 3137, average=6.16), Pvalue = 0.93

(b) Village Mortality Info Treatment versus Control



Note: Treatment (N:2264, average=6.19), Control: (N : 2229, average=6.28), Pvalue = 0.48

Notes: The figure shows a Kernel estimate of the distribution of the gap between the perceived rate reported by women at endline and the actual, local mortality rate measured during the listing for children born in 2007-2012. Graph (a) restricts the sample to villages assigned to individual interventions and compares the distribution in the mortality information treatment arm (in gray) and the distribution in the pure control arm (in black). Graph (b) restricts the sample to villages assigned to debates and compares the distribution in the debate + mortality information treatment arm (in gray) and the distribution in the pure debate arm (in black). The solid vertical lines indicate the average in each group; the dashed vertical lines indicate the first and third quartiles in each group. Rates are expressed in percentage points.

Table A.1: Balancing tests in baseline survey (for the non-attrition sample)

	Baseline Mean		(3) N	Difference	
	(1) Control	(2) Treatment		(4) Effect Size	(5) P-value
# of pregnancies before baseline	3.493	3.525	12546	.007	0.76
Currently using modern contraception	.332	.312	12546	-.03	0.34
Total # of children desired	5.4	5.479	12546	.039	0.31
Total # of children desired (Husband)	5.564	5.637	12546	.028	0.47
Under-5 mortality rate (%)	14.471	16.289	12546	.097	0.05
Agrees: there is a quantity-quality tradeoff	.712	.690	12546	-.086	0.22
Agrees: times are changing and there is no social norm on family size	.754	.727	12546	-.098	0.21
Agrees: modern contraception is a reliable way to control births	.793	.78	12546	-.065	0.45
Agrees: there is a quantity-quality tradeoff	.468	.444	12546	-.067	0.20
Agrees: times are changing and there is no social norm on family size	.504	.479	12546	-.069	0.21
Has unmet need for contraception	.378	.391	12540	.016	0.60
Wife does not want another child in next 2 years	.656	.655	11968	-.014	0.60
Husband does not want another child in next 2 years	.579	.5730	8505	-.024	0.34
Currently using modern contraception	.332	.313	12537	-.029	0.34
At baseline: not using modern contraception and not pregnant	.578	.587	12546	.01	0.74
Could not afford contraception if ever wanted to use it	.386	.436	11458	.073	0.03
Agrees: modern contraception is not dangerous to health	.782	.796	11258	.016	0.56
Wife does not want another child in next 2 years	.656	.655	11968	-.014	0.60
Agrees: community disapproves couple using contraception to delay 1st birth	.645	.648	12546	.035	0.23
Reports women sometimes punished or stigmatized for using contraception	.398	.367	11551	-.046	0.14
Wife's age	28.359	28.372	12546	.013	0.59
Husband could be surveyed	.759	.742	12546	-.02	0.64
Age Gap between Husband and Wife	9.791	10.2	12546	.035	0.33
Husband's age at baseline missing	.035	.032	12546	-.015	0.52
Wife reports husband is polygamous	.428	.46	12546	.048	0.24
Ever attended formal school?	.178	.161	12546	-.026	0.45
Has had at least one child who died	.276	.267	12546	-.023	0.35
HH has a radio	.478	.496	12546	.062	0.09
Municipality covered by DMI	.577	.565	12546	-.04	0.75
Equals 1 if radio that broadcasts the DMI PF ads, 0 otherwise	.331	.379	12546	.051	0.67
Wants another child	.924	.921	12546	-.007	0.72
Wants another child in next 2 years	.344	.345	11968	.014	0.60
Ever used modern contraception	.478	.463	12539	-.012	0.70
Currently using modern contraception	.332	.313	12537	-.029	0.34
Wife overestimates under 5 child mortality	.521	.579	9717	.102	0.01
Health Center < 2km away	.166	.17	12546	.034	0.59
Agrees: modern contraception is a reliable way to control births	.513	.502	12546	-.042	0.45

Notes: columns 1 and 2 display the mean of each variable in the 100% subsidy (treatment) and in the 10% subsidy arm (control). To compute the effect size (column 4) and the p-value (column 5), all specifications include province fixed effects and endline controls (whether the village had to be surveyed by phone due to security concerns at endline and date of the endline survey). Robust standard errors are clustered at health center level. The sample is restricted to individuals surveyed at endline (the analytic sample).

Table A.2: Baseline visit outcomes (for the non-attrition sample)

	(1)	(2)	(3)	(4)
	Accepted voucher booklet	Refused voucher because no need	Refused voucher because scared husband/ someone finds out	Husband could be surveyed
Full Subsidy	0.043** (0.021)	-0.020 (0.015)	-0.008 (0.006)	-0.009 (0.018)
Province FE	Yes	Yes	Yes	Yes
Baseline Controls	No	No	No	No
Observations	12,546	12,529	12,529	12,546
Control (10% Subsidy) Mean	0.679	0.145	0.056	0.746

Notes: All specifications include province fixed effects. Controls are: whether the village had to be surveyed by phone due to security concerns at endline and date of the endline survey. Robust standard errors in parentheses. Clustering at health center level. The sample is restricted to individuals surveyed at endline (the analytic sample). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.3: Attrition at endline

	(1)	(2)
	Focal wife surveyed	Husband surveyed
Full Subsidy	-0.016 (0.016)	-0.012 (0.016)
Baseline Controls	No	No
Observations	14,609	14,609
Control (10% Subsidy) Mean	0.874	0.884

Notes: All specifications include Province fixed effects. Robust standard errors in parentheses. Clustering at health center level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.4: Summary statistics from baseline survey: husbands

	Mean	SD	N
Husband's age	40.24	11.37	10,781
Focal wife's age	28.58	5.42	10,782
Wife reports husband is polygamous	0.48	0.50	10,784
Husband has no formal education	0.81	0.40	10,784
Muslim	0.64	0.48	10,778
Fertility:			
# of children (from all wives)	6.06	4.91	8,567
Wants another child	0.92	0.27	10,437
Wants another child in next 2 years	0.44	0.50	9,734
Total # of children desired	9.48	6.98	8,527
Exposure to contraception:			
Ever heard of contraception/methods to delay births	0.89	0.31	10,778
Ever used modern contraception	0.40	0.49	10,504
Currently using modern contraception	0.31	0.46	10,178
Distance to closest local health center (kilometres)	5.42	4.06	10,784
Personal views (first-order beliefs):			
Agrees: modern contraception is not dangerous to health	0.74	0.44	9,303
Agrees: modern contraception is not against tradition	0.78	0.41	9,346
Agrees: modern contraception is a reliable way to control births	0.90	0.31	8,056
Agrees: there is a quantity-quality tradeoff	0.80	0.40	8,097
Perceived social norms (second-order beliefs):			
Agrees: times are changing and there is no social norm on family size	0.86	0.35	8,149
Agrees: community disapproves use of contraception to delay 1st birth	0.61	0.49	10,784
Reports women sometimes punished/stigmatized for using contraception	0.37	0.48	9,636
Child mortality:			
True under 5 mortality rate (%)	10.56	2.12	10,784
Husband's perceived under 5 mortality rate (%)	17.63	16.73	9,651
Husband overestimates under-5 mortality	0.52	0.50	9,651

Notes: Data from Baseline survey with husbands.

Table A.5: Treatment effects: other potential benefits to the wife

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	Emotional Violence Index (12m)	Physical Violence Index (12m)	Sexual Violence Index (12m)	IPV Index (past 12m)	Self reported health today is very good	Agrees to statement that life is good	Husband polyga- mous at endline	Satisfied with using birth control method	Should a woman have control over the number of children?	Agrees that contraception allows households greater control over their lives	Has no one to talk to about reproductive health
Full Subsidy	-0.043 (0.036)	-0.016 (0.028)	-0.057** (0.027)	-0.040** (0.020)	0.037 (0.027)	-0.026 (0.023)	-0.001 (0.009)	0.011 (0.009)	-0.037 (0.024)	-0.006 (0.033)	-0.019 (0.016)
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Baseline Controls	No	No	No	No	No	No	Yes	No	No	No	No
Observations	8,399	8,327	8,400	8,402	12,527	12,499	11,774	5,966	8,295	7,934	12,038
Control (10% Subsidy) Mean	0.001	0.000	0.000	0.009	0.395	0.759	0.430	0.902	0.455	0.421	0.143

Notes: In the endline survey, we included a module on intimate partner violence (IPV). We used well-validated questions to measure various domains of violence, and extensively piloted them in the field during summer 2019. The sample size for the IPV module is smaller because it could only be administered in-person; for women surveyed by phone, we could not guarantee confidentiality. Endline controls are: Whether the village had to be surveyed by phone due to security concerns at endline and date of the endline survey. Province fixed effects are used in all specifications. Robust standard errors in parentheses. Clustering at the health center level. Column 7 additionally controls for husband's polygamous status at baseline. ***, $p < 0.01$, **, $p < 0.05$, *, $p < 0.1$

Table A.6: Treatment effects on primary outcomes by subsamples (with baseline controls)

		(1)	(2)	(3)	(4)	(5)
Sub-sample		Had a live birth since baseline	Had a pregnancy since baseline	Used medical contraception in last 3 yrs	Month(s) used modern contraception (last spell of each type) in last 3 yrs	Used IPA subsidy voucher
Need for Contraception:						
Had unmet need for contraception at baseline (N=4,649)	Full Subsidy	-0.011 (0.014)	-0.014 (0.014)	0.022 (0.018)	-0.062 (0.475)	0.043*** (0.014)
	Control Mean	0.662	0.736	0.437	7.229	0.128
Wife did not want another child over next 2 yrs at baseline (N=7,583)	Full Subsidy	-0.019 (0.013)	-0.024* (0.013)	0.020 (0.015)	-0.166 (0.428)	0.040*** (0.012)
	Control Mean	0.611	0.703	0.559	10.649	0.157
Husband did not want another child over next 2 yrs at baseline (N=4,724)	Full Subsidy	-0.014 (0.016)	-0.035** (0.016)	0.031* (0.017)	0.311 (0.448)	0.036*** (0.012)
	Control Mean	0.612	0.712	0.560	10.594	0.155
Was not using modern contraception at baseline (N=8,191)	Full Subsidy	-0.010 (0.012)	-0.012 (0.011)	0.000 (0.014)	-0.283 (0.362)	0.030*** (0.009)
	Control Mean	0.641	0.709	0.421	6.854	0.113
Was not using modern contraception and was not pregnant at baseline (N=7,052)	Full Subsidy	-0.014 (0.013)	-0.013 (0.013)	0.005 (0.015)	-0.014 (0.384)	0.034*** (0.009)
	Control Mean	0.585	0.677	0.408	6.551	0.109
Could not afford contraception at baseline (N=4,519)	Full Subsidy	-0.027* (0.016)	-0.034** (0.014)	0.014 (0.016)	-0.009 (0.498)	0.039*** (0.013)
	Control Mean	0.616	0.702	0.505	9.055	0.148
Other Frictions:						
Health Center < 2km away (N=2,060)	Full Subsidy	-0.025 (0.028)	-0.037 (0.028)	0.055* (0.030)	0.669 (0.834)	0.088*** (0.022)
	Control Mean	0.578	0.680	0.558	10.022	0.150
Disagrees with modern contraception being harmful for health (N=8,563)	Full Subsidy	-0.008 (0.012)	-0.012 (0.013)	0.002 (0.014)	-0.481 (0.391)	0.038*** (0.010)
	Control Mean	0.617	0.705	0.573	10.633	0.159
Neither wanted another child over next 2 years (N=3,771)	Full Subsidy	-0.013 (0.018)	-0.037** (0.018)	0.039** (0.019)	0.318 (0.481)	0.037** (0.014)
	Control Mean	0.599	0.709	0.567	11.044	0.163

Notes: Endline controls are: Whether the village had to be surveyed by phone due to security concerns at endline and date of the endline survey. Province fixed effects are used in all specifications. Robust standard errors in parentheses. Clustering at health center level. Control Mean refers to that of 10% subsidy sub-sample. Province fixed effects used across all specifications. The median age of wives is 28. The regions with family planning pilot in 2019 are Cascades and Centre Ouest. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.7: Treatment effects on primary outcomes: other subsamples

		(1)	(2)	(3)	(4)	(5)
Sub-sample		Had a live birth since baseline	Had a pregnancy since baseline	Used medical contraception in last 3 yrs	Month(s) used modern contraception (last spell of each type) in last 3 yrs	Used IPA subsidy voucher
Young wives ($\leq$ median age)	Full Subsidy	-0.020 (0.015)	-0.013 (0.013)	0.016 (0.016)	-0.266 (0.474)	0.021* (0.012)
	Observations	6,290	6,291	6,066	6,056	6,277
	Control Mean	0.722	0.810	0.549	9.255	0.148
Older wives ($>$ median age)	Full Subsidy	-0.011 (0.014)	-0.022 (0.016)	-0.016 (0.018)	-0.419 (0.443)	0.044*** (0.011)
	Observations	6,252	6,252	6,065	6,051	6,242
	Control Mean	0.523	0.599	0.512	9.962	0.136
Monogamous husband	Full Subsidy	-0.018 (0.013)	-0.018 (0.013)	0.010 (0.016)	-0.077 (0.420)	0.031*** (0.011)
	Observations	6,974	6,975	6,756	6,743	6,969
	Control Mean	0.660	0.750	0.556	9.870	0.149
Polygamous husband	Full Subsidy	-0.011 (0.014)	-0.015 (0.013)	-0.010 (0.018)	-0.635 (0.481)	0.034*** (0.012)
	Observations	5,568	5,568	5,375	5,364	5,550
	Control Mean	0.573	0.645	0.496	9.256	0.132
Senior wife in a polygamous marriage	Full Subsidy	-0.014 (0.025)	-0.013 (0.025)	-0.012 (0.029)	-0.741 (0.675)	0.050*** (0.019)
	Observations	1,728	1,728	1,669	1,666	1,721
	Control Mean	0.556	0.626	0.515	9.971	0.126
Junior wife in a polygamous marriage	Full Subsidy	-0.009 (0.016)	-0.016 (0.015)	-0.011 (0.019)	-0.576 (0.532)	0.026** (0.013)
	Observations	3,840	3,840	3,706	3,698	3,829
	Control Mean	0.581	0.653	0.487	8.937	0.134
Low # of pregnancies at baseline (< 5)	Full Subsidy	-0.013 (0.013)	-0.016 (0.012)	0.002 (0.015)	-0.336 (0.432)	0.024** (0.011)
	Observations	8,927	8,928	8,622	8,608	8,909
	Control Mean	0.679	0.769	0.523	8.977	0.140
High # of pregnancies at baseline (≥ 5)	Full Subsidy	-0.027 (0.018)	-0.027 (0.020)	-0.007 (0.021)	-0.345 (0.527)	0.054*** (0.013)
	Observations	3,615	3,615	3,509	3,499	3,610
	Control Mean	0.483	0.545	0.550	11.173	0.146
Was using modern contraception at baseline	Full Subsidy	-0.028 (0.018)	-0.027 (0.019)	0.014 (0.019)	-0.267 (0.609)	0.041*** (0.014)
	Observations	4,043	4,043	3,911	3,907	4,037
	Control Mean	0.587	0.697	0.752	15.159	0.200
Excluding regions with Family Planning pilot (2019)	Full Subsidy	-0.020 (0.012)	-0.022* (0.013)	-0.003 (0.016)	-0.372 (0.424)	0.032*** (0.010)
	Observations	11,378	11,379	11,014	10,991	11,357
	Control Mean	0.623	0.706	0.548	9.936	0.147
Excluding villages affected by insecurity	Full Subsidy	-0.020 (0.013)	-0.016 (0.014)	0.023 (0.017)	0.373 (0.415)	0.049*** (0.011)
	Observations	8,280	8,280	8,263	8,251	8,275
	Control Mean	0.601	0.680	0.496	8.842	0.109

Notes: Endline controls are: Whether the village had to be surveyed by phone due to security concerns at endline and date of the endline survey. Province fixed effects are used in all specifications. Robust standard errors in parentheses. Clustering at health center level. Control Mean refers to that of 10% subsidy sub-sample. Province fixed effects used across all specifications. The median age of wives is 28. The regions with family planning pilot in 2019 are Cascades and Centre Ouest. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.8: Treatment effects when potential misperceptions are absent (with baseline controls)

		(1)	(2)	(3)	(4)	(5)
Sub-sample		Had a live birth since baseline	Had a pregnancy since baseline	Used medical contraception in last 3 yrs	Month(s) used modern contraception (last spell of each type) in last 3 yrs	Used IPA subsidy voucher
Social Norms:						
Does not believe community disapproves use of contraception (N=4,277)	Full Subsidy	-0.020 (0.014)	-0.019 (0.013)	0.003 (0.016)	-0.568 (0.477)	0.044*** (0.013)
	Control Mean	0.627	0.703	0.531	9.600	0.150
Does not report women being punished or stigmatized for using contraception (N=6,855)	Full Subsidy	-0.023 (0.014)	-0.021 (0.014)	-0.005 (0.014)	-0.531 (0.426)	0.035*** (0.011)
	Control Mean	0.634	0.714	0.559	10.239	0.155
Mortality Perceptions:						
Does not overestimate under-5 child mortality (N=4,227)	Full Subsidy	-0.012 (0.016)	-0.001 (0.016)	0.034** (0.015)	0.433 (0.451)	0.035*** (0.012)
	Control Mean	0.628	0.707	0.527	9.473	0.148

Notes: Endline controls are: Whether the village had to be surveyed by phone due to security concerns at endline and date of the endline survey. Province fixed effects are used in all specifications. Robust standard errors in parentheses. Clustering at health center level. Control Mean refers to that of 10% subsidy sub-sample. Province fixed effects used across all specifications. The median age of wives is 28. The regions with family planning pilot in 2019 are Cascades and Centre Ouest. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.9: Treatment effects on primary outcomes: fully interacted (with baseline controls)

	(1)	(2)	(3)	(4)	(5)
	Had a live birth since baseline	Had a pregnancy since baseline	Used medical contraception in last 3 yrs	Month(s) used modern contraception (last spell of each type) in last 3 years	Used IPA subsidy voucher
Full Subsidy	-0.034* (0.018)	-0.024 (0.018)	0.013 (0.021)	-0.196 (0.475)	0.050*** (0.016)
Village Debate or Edutainment	-0.015 (0.016)	-0.002 (0.015)	0.014 (0.018)	0.293 (0.455)	0.034** (0.013)
Individual Edutainment	0.009 (0.021)	0.006 (0.020)	0.000 (0.021)	0.480 (0.623)	0.016 (0.016)
Individual Mortality Info	-0.000 (0.023)	-0.004 (0.019)	0.021 (0.022)	0.800 (0.641)	-0.014 (0.014)
Village Interventions X Full Subsidy	0.022 (0.022)	0.003 (0.021)	-0.007 (0.025)	0.236 (0.599)	-0.024 (0.020)
Individual Edutainment X Full Subsidy	0.027 (0.030)	0.039 (0.029)	0.021 (0.030)	-0.406 (0.804)	-0.026 (0.026)
Individual Mortality Info X Full Subsidy	0.025 (0.033)	0.006 (0.030)	-0.030 (0.032)	-0.800 (0.829)	0.015 (0.022)
Province FE	Yes	Yes	Yes	Yes	Yes
Baseline Controls	Yes	Yes	Yes	Yes	Yes
Observations	12,542	12,543	12,131	12,107	12,519
Control Mean	0.628	0.701	0.526	9.373	0.129

Notes: Province fixed effects are used in all specifications. Robust standard errors in parentheses. Clustering at the village level. Control mean refers to that of the pure control. Baseline controls are: number of births at baseline; whether wife was using modern contraception at baseline; number of children desired by wife at baseline ; number of children desired by husband at baseline ;under 5 mortality rate reported by wife at baseline; wife and husband's first order beliefs at baseline i.e. whether or not they each agree that "there is a quantity-quality trade-off", "times are changing and there is no social norm on family size" and (in response to a vignette) "Z. should use long-lasting contraception to delay 5th birth"; wife's age at baseline, spousal age gap at baseline, polygamous union at baseline, whether husband was surveyed at baseline, whether the wife has ever gone to school, whether she has had a child who died at baseline, exposure to DMI radio programs at baseline. Endline controls are: Whether the village had to be surveyed by phone due to security concerns at endline and date of the endline survey. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Table A.10: Treatment effects of village interventions on wife's beliefs

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	First-order beliefs				Second-order beliefs			
	Modern contracep- tion is not dangerous to health	Modern contracep- tion is not against tradition	Modern contracep- tion is a reliable way to control births	There is a quantity- quality tradeoff	Community disapproves couple using contraception to delay 1st birth	Women sometimes punished/ stigmatized for using contraception	Modern contracep- tion is a reliable way to control births	There is a quantity- quality tradeoff
Panel A: Effect of village interventions								
Village Debate or Edutainment	0.001 (0.010)	-0.029*** (0.011)	-0.008 (0.006)	-0.007 (0.010)	0.010 (0.015)	0.011 (0.013)	0.000 (0.004)	0.002 (0.004)
CSPS FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Baseline Controls	No	No	No	No	No	No	No	No
Observations	11,459	11,461	12,546	12,546	12,546	11,314	12,511	12,501
Control Mean	0.818	0.814	0.908	0.841	0.649	0.483	0.627	0.524
Panel B: Effect of village interventions by baseline beliefs								
Village Debate or Edutainment	-0.002 (0.018)	-0.008 (0.019)	-0.022 (0.028)	-0.014 (0.022)	0.001 (0.019)	0.008 (0.017)		
Agrees at baseline	0.051*** (0.014)	0.014 (0.015)	0.068*** (0.018)	0.073*** (0.015)	0.018 (0.012)	0.030** (0.014)		
Village Int. X Agrees at baseline	0.004 (0.019)	-0.023 (0.020)	0.012 (0.028)	0.010 (0.022)	0.014 (0.017)	-0.001 (0.020)		
CSPS FE	Yes	Yes	Yes	Yes	Yes	Yes		
Baseline Controls	No	No	No	No	No	No		
Observations	10,331	10,358	10,588	10,498	12,546	10,464		
Control Mean	0.775	0.822	0.826	0.811	0.627	0.478		

Notes: Robust standard errors in parentheses. Clustering at village level. CSPS fixed effects used across all specifications. Endline controls used are: Whether the village had to be surveyed by phone due to security concerns at endline and date of the endline survey. In all regressions, we include other demand treatment arms (Individual Mortality Info and Individual Edutainment) but we don't report the coefficients in the table. The control mean in Panel A is the average outcome in the group that did not receive any demand intervention. The control mean in Panel B is the average outcome in the group that did not receive any demand intervention and disagrees at baseline. Second order beliefs in column 7 and 8 refer to the proportion of people in the community that the respondents think agree with the outcome statement as opposed to columns 3 and 4 where the outcome measures whether the respondent herself agrees with the outcome statement. Panel B controls for whether the wife agreed with the outcome statement at baseline and interacts this with the village interventions. We cannot estimate the regression in columns 7 and 8 in panel B because we did not collect the relevant variable at baseline. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.11: Treatment effects of village interventions on primary outcomes by baseline beliefs

		(1)	(2)	(3)	(4)	(5)
Sub-sample		Had a live birth since baseline	Had a pregnancy since baseline	Used medical contraception in last 3 yrs	Month(s) used modern contraception (last spell of each type) in last 3 yrs	Used IPA subsidy voucher
<u>Personal views (first-order beliefs):</u>						
Modern contraception is not dangerous to health (N=10867)	Village Debate or Edutainment	-0.037* (0.020)	-0.016 (0.019)	0.009 (0.022)	0.295 (0.541)	-0.007 (0.016)
	Agrees at baseline	-0.058*** (0.016)	-0.031** (0.016)	0.077*** (0.018)	1.790*** (0.417)	0.025** (0.013)
	Village Interventions X Agrees at baseline	0.042* (0.022)	0.023 (0.021)	-0.011 (0.023)	0.114 (0.584)	0.028* (0.016)
Modern contraception is not against tradition (N=10900)	Village Debate or Edutainment	-0.017 (0.021)	-0.008 (0.022)	0.022 (0.022)	0.451 (0.558)	-0.003 (0.017)
	Agrees at baseline	-0.018 (0.018)	-0.014 (0.017)	0.030 (0.019)	0.376 (0.463)	-0.012 (0.012)
	Village Interventions X Agrees at baseline	0.018 (0.024)	0.016 (0.023)	-0.027 (0.024)	-0.125 (0.595)	0.021 (0.017)
Modern contraception is a reliable way to control births (N=10227)	Village Debate or Edutainment	0.022 (0.038)	0.041 (0.036)	-0.020 (0.038)	-0.916 (0.911)	0.033 (0.022)
	Agrees at baseline	0.032 (0.025)	0.049** (0.024)	0.122*** (0.025)	1.213** (0.609)	0.076*** (0.014)
	Village Interventions X Agrees at baseline	-0.036 (0.039)	-0.050 (0.036)	0.034 (0.038)	1.711* (0.894)	-0.006 (0.021)
There is a quantity-quality tradeoff (N=10138)	Village Debate or Edutainment	-0.033 (0.025)	-0.022 (0.023)	0.013 (0.027)	0.751 (0.639)	0.013 (0.020)
	Agrees at baseline	-0.016 (0.018)	-0.017 (0.016)	0.084*** (0.019)	1.892*** (0.442)	0.017 (0.014)
	Village Interventions X Agrees at baseline	0.028 (0.026)	0.021 (0.024)	0.003 (0.027)	-0.028 (0.638)	0.017 (0.020)
<u>Perceived social norms (second-order beliefs):</u>						
There is no social norm on family size (N=10275)	Village Debate or Edutainment	0.005 (0.028)	0.013 (0.027)	-0.036 (0.030)	-0.600 (0.708)	0.013 (0.021)
	Agrees at baseline	0.014 (0.021)	0.024 (0.020)	0.053** (0.021)	0.899* (0.539)	0.031** (0.015)
	Village Interventions X Agrees at baseline	-0.017 (0.029)	-0.019 (0.028)	0.059** (0.030)	1.527** (0.730)	0.017 (0.021)
Community disapproves couple using contraception to delay 1st birth (N=12107)	Village Debate or Edutainment	-0.008 (0.016)	0.014 (0.014)	0.007 (0.017)	0.730* (0.396)	0.008 (0.014)
	Agrees at baseline	0.000 (0.013)	0.020 (0.013)	0.014 (0.014)	0.607* (0.330)	-0.011 (0.010)
	Village Interventions X Agrees at baseline	0.010 (0.018)	-0.012 (0.017)	-0.005 (0.019)	-0.554 (0.444)	0.008 (0.014)
Reports women sometimes punished/stigmatized for using contraception (N=11152)	Village Debate or Edutainment	-0.002 (0.013)	0.011 (0.012)	0.005 (0.015)	0.204 (0.370)	0.007 (0.012)
	Agrees at baseline	0.008 (0.014)	0.022* (0.013)	0.004 (0.014)	-0.355 (0.352)	-0.018* (0.011)
	Village Interventions X Agrees at baseline	-0.002 (0.019)	-0.018 (0.017)	-0.009 (0.020)	0.475 (0.488)	0.021 (0.015)

Notes: Robust standard errors in parentheses. Clustering at village level. CSPS fixed effects used across all specifications. Endline controls used are: Whether the village had to be surveyed by phone due to security concerns at endline and date of the endline survey. In all regressions, we include other demand treatment arms (Individual Mortality Info and Individual Edutainment) but we don't report the coefficients in the table. N refers to the number of observations that had a non-missing response to the belief question at baseline. The number of observations may vary slightly between outcomes; in this case, we report the minimum number across all 4 regressions.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A.12: High returns to quantity

Statistic	Mean	N	Mean	N	Source
	Husband		Wife		
Will not be able to cope in old age without support of child	0.74	206	0.81	226	Scoping
Share of children expected to send enough money back to sustain parents	0.20	252	0.26	277	Scoping
Labor constrained to expand farm activity	0.39	67523			Listing
Associate lack of children with: No labour for land	0.10	8567	0.13	12424	Baseline
Associate lack of children with: No labour for chores	0.07	8567	0.17	12424	Baseline
Associate lack of children with: Unhappiness	0.07	8567	0.14	12424	Baseline

Notes: Scoping visits were conducted between September 2016 and March 2017. We conducted semi-qualitative interviews with married men and women of reproductive age across 8 regions. The listing exercise took place in Fall 2017 in 500 villages. The baseline took place in Spring 2018 in 499 villages.

Appendix B

Table B.1: Treatment effects: spousal communication

	(1)	(2)	(3)	(4)
	Wife		Husband	
	Talked to husband about modern contraception in past 3 years	Ever discussed number of children with husband	# wives with whom husband ever talked about contraception	# wives with whom husband ever talked about number of children
Full Subsidy	0.010 (0.017)	-0.036** (0.016)	-0.063** (0.026)	-0.052*** (0.019)
Province FE	Yes	Yes	Yes	Yes
Baseline Controls	No	No	No	No
Observations	11,440	12,535	11,774	11,774
Control (10% Subsidy) Mean	0.588	0.320	0.726	0.335

Notes: Endline controls are: Whether the village had to be surveyed by phone due to security concerns at endline and date of the endline survey. Robust standard errors in parentheses. Clustering at health center level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.2: Treatment effects on primary outcomes: fully interacted (version with village interventions not pooled together)

	(1)	(2)	(3)	(4)	(5)
	Had a live birth since baseline	Had a pregnancy since baseline	Used medical contraception in last 3 yrs	Month(s) used modern contraception (last spell of each type) in last 3 years	Used IPA subsidy voucher
Full Subsidy	-0.036* (0.019)	-0.025 (0.018)	0.006 (0.023)	-0.281 (0.541)	0.047*** (0.017)
Village Debate	-0.008 (0.021)	0.015 (0.020)	0.008 (0.025)	0.031 (0.661)	0.018 (0.017)
Village Debate + Mortality Info	-0.017 (0.021)	0.003 (0.019)	0.009 (0.029)	0.282 (0.703)	0.031 (0.019)
Village Edutainment	-0.007 (0.021)	0.006 (0.020)	0.018 (0.025)	0.552 (0.630)	0.029** (0.015)
Individual Edutainment	0.006 (0.022)	0.003 (0.020)	-0.006 (0.021)	0.393 (0.627)	0.013 (0.015)
Individual Mortality Info	0.010 (0.025)	0.008 (0.021)	0.025 (0.022)	0.886 (0.665)	-0.015 (0.014)
Village Debate X Full Subsidy	0.033 (0.030)	0.000 (0.029)	0.008 (0.037)	1.068 (0.924)	-0.023 (0.026)
Village Debate + Mortality Info X Full Subsidy	0.013 (0.030)	-0.011 (0.028)	-0.031 (0.041)	-0.520 (0.954)	-0.021 (0.028)
Village Edutainment X Full Subsidy	0.021 (0.028)	0.014 (0.029)	-0.012 (0.037)	-0.232 (0.895)	-0.036 (0.026)
Individual Edutainment X Full Subsidy	0.031 (0.031)	0.044 (0.030)	0.030 (0.031)	-0.285 (0.832)	-0.024 (0.025)
Individual Mortality Info X Full Subsidy	0.027 (0.034)	0.009 (0.031)	-0.025 (0.033)	-0.728 (0.842)	0.017 (0.022)
Province FE	Yes	Yes	Yes	Yes	Yes
Baseline Controls	No	No	No	No	No
Observations	12,542	12,543	12,131	12,107	12,519
Control (10% Subsidy) Mean	0.628	0.701	0.526	9.373	0.129

Notes: See [Table 6](#).

Table B.3: Treatment effects on primary outcomes

	(1)	(2)	(3)	(4)	(5)
	Had a live birth since baseline	Had a pregnancy since baseline	Used medical contraception in last 3 yrs	Month(s) used modern contraception (last spell of each type) in last 3 years	Used IPA subsidy voucher
<u>Panel A: Supply Intervention (Price Subsidy)</u>					
Full Subsidy	-0.017 (0.012)	-0.019 (0.012)	0.000 (0.015)	-0.318 (0.399)	0.032*** (0.010)
Province FE	Yes	Yes	Yes	Yes	Yes
Baseline Controls	No	No	No	No	No
Observations	12,542	12,543	12,131	12,107	12,519
Control (10% Subsidy) Mean	0.623	0.705	0.531	9.609	0.142
<u>Panel B: Demand Interventions: Pooled</u>					
Village Debate or Edutainment	-0.002 (0.010)	0.006 (0.010)	0.003 (0.013)	0.371 (0.307)	0.013 (0.010)
Individual Edutainment	0.021 (0.015)	0.025 (0.015)	0.009 (0.015)	0.228 (0.416)	0.003 (0.012)
Individual Mortality Info	0.022 (0.017)	0.013 (0.015)	0.015 (0.016)	0.562 (0.420)	-0.004 (0.011)
CSPS FE	Yes	Yes	Yes	Yes	Yes
Baseline Controls	No	No	No	No	No
Edutain. Indiv=Vill.	0.108	0.184	0.727	0.732	0.412
<u>Panel C: Demand Interventions: Breakdown</u>					
Village Debate	0.007 (0.014)	0.012 (0.013)	0.011 (0.016)	0.661 (0.411)	0.007 (0.012)
Village Debate + Mortality Info	-0.012 (0.014)	-0.005 (0.012)	-0.014 (0.017)	-0.110 (0.405)	0.021 (0.013)
Village Edutainment	0.000 (0.013)	0.011 (0.012)	0.013 (0.017)	0.541 (0.395)	0.012 (0.012)
Individual Edutainment	0.021 (0.015)	0.025 (0.015)	0.009 (0.015)	0.227 (0.416)	0.003 (0.012)
CSPS FE	Yes	Yes	Yes	Yes	Yes
Baseline Controls	No	No	No	No	No
Edutain. Indiv=Vill.	0.198	0.372	0.810	0.516	0.529
Deb=Deb + Mortality	0.207	0.221	0.180	0.096	0.324
Deb = Edutain. Vil	0.662	0.905	0.885	0.793	0.705

Notes: All specifications include province fixed effects (Panel A) or health centers (CSPS) fixed effects (Panels B and C). Controls are: Whether the village had to be surveyed by phone due to security concerns at endline and date of the endline survey. Robust standard errors in parentheses. Panel A: clustering at health center level. Panels B and C: clustering at village level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table B.4: Treatment effects of demand interventions on perceived Under 5 mortality rates, by sub-samples

	(1) Women who Over- estimate	(2) Women who Underesti- mate	(3) Women who had lost child	(4) Women who had not lost child	(5) Province Where Mortality Increased	(6) Province Where Mortality Decreased
Village Debate	0.340 (0.685)	-0.624 (0.601)	0.606 (0.909)	-0.146 (0.503)	-0.699 (0.671)	0.171 (0.630)
Village Debate + Mortality Info	0.649 (0.647)	-0.809 (0.594)	1.596* (0.860)	-0.285 (0.476)	1.108 (0.741)	-0.691 (0.531)
Village Edutainment	0.595 (0.639)	-1.344** (0.633)	-1.170 (0.883)	-0.161 (0.468)	-1.048 (0.673)	-0.238 (0.559)
Individual Edutainment	0.065 (0.645)	0.129 (0.731)	2.256** (1.106)	-0.167 (0.466)	1.231* (0.741)	-0.415 (0.467)
Individual Mortality Info	-0.130 (0.695)	-0.822 (0.708)	0.413 (0.961)	-0.475 (0.475)	0.286 (0.768)	-0.439 (0.493)
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
Baseline Controls	No	No	No	No	No	No
Observations	5,299	4,314	2,430	9,039	3,276	9,117
Control (10% Subsidy) Mean	16.912	16.199	16.687	16.318	16.780	16.314

Notes: The table presents the effects of demand interventions on perceived under-5 mortality rates for different sub-samples: women who overestimated (col 1) or underestimated (col 2) child mortality at baseline; women who had lost (col 3) or not (col 4) a child at baseline; provinces where the mortality increased (col 5) or decreased (col 6) between baseline and endline. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Appendix C: Impact of macro shocks

Several events took place during our study period: namely the gradual deterioration of security, the health sector strike between July 2019 and December 2019, the onset of COVID in March 2020, and finally the announcement of the national FP policy in July 2020. We now discuss whether our null effects could be explained by these events and the extent to which they may have modified the impact of the free subsidy. We start by noting that the effect of these events on births is uncertain. For example COVID could disrupt access to contraception, but also temporarily lower the demand for children given the new health risks.

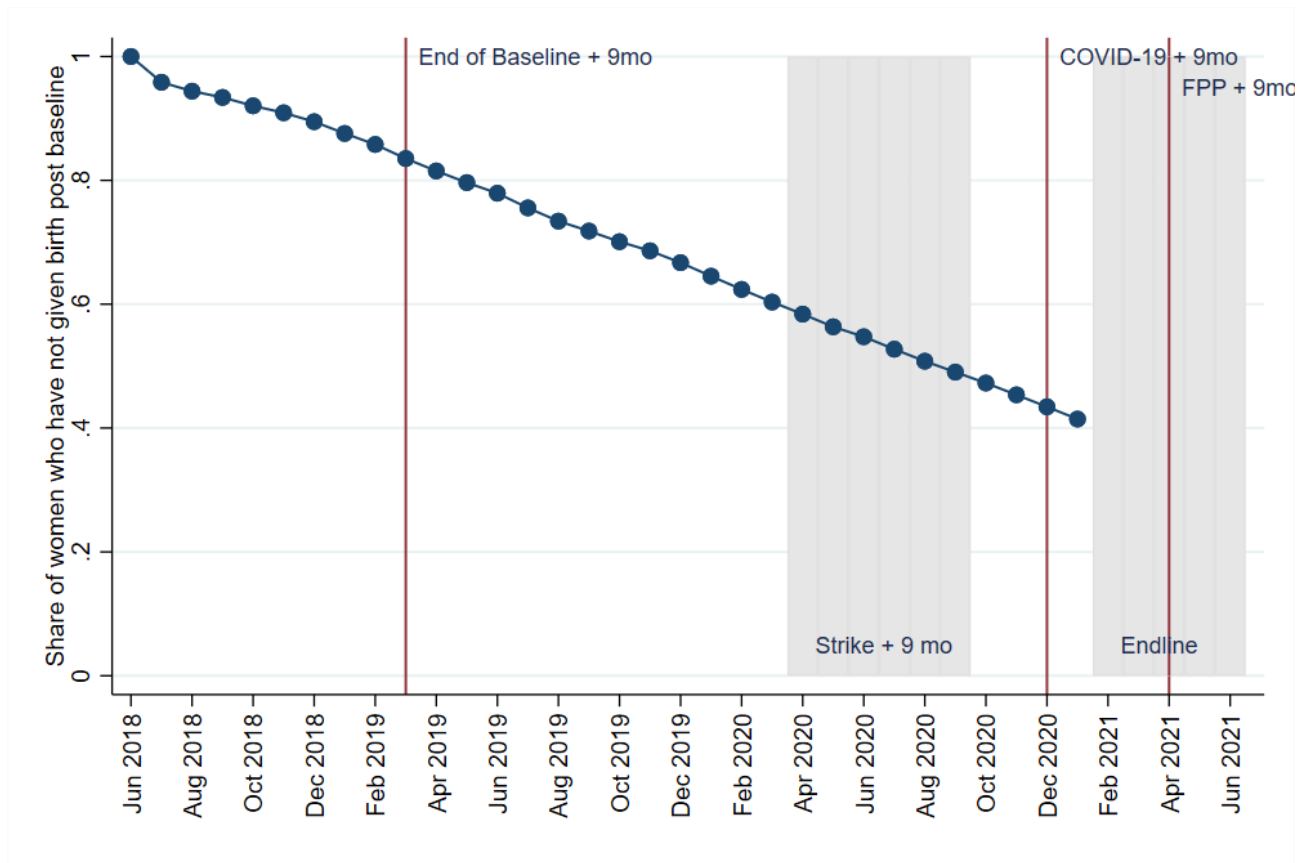
We start by showing in [Figure C.1](#) how the timing of these events appears to affect the share of women who have not given birth in our sample, following them month by month. First, COVID and the national FP policy happened too late to be a threat; these events could only affect births towards the very end of our study period. Second, we see that the likelihood of having a child varies smoothly throughout the study period; in particular there is no change in slope during the period corresponding to 9 months after the strike, suggesting that the strike is unlikely to have caused a sudden shortage of contraceptives and a rise in unwanted pregnancies.

To investigate the effect of security issues, we exclude health centers strongly affected by the violence (using our monitoring data). The results are reported in [Table A.7](#). The estimates remain small and insignificant. Finally, the government made contraception free ahead of the roll-out of its national program in two regions, Cascades and Centre Ouest (see [Figure A.3](#)). [Table A.7](#) shows that the null results are robust to excluding these two regions.

We conclude that our null results are not explained by the fact that the intervention took place during tumultuous times. We also note that times are in fact often tumultuous in this region, which makes our results policy-relevant precisely because we were able to implement the experiment during these times.³⁸

³⁸Beyond Burkina Faso, the jihadist violence has been affecting Mali, Niger, Mauritania, Chad, Cameroon, Benin, Togo and Cote d'Ivoire for more than 10 years. Just after our endline, there were two military coups in Burkina Faso (January 2022 and September 2022). Between 2020 and 2024, there were military coups in Mali, Chad, Guinea, Sudan, Niger and Gabon.

Figure C.1: Share of women who had not given birth by calendar month



Note: The baseline and interventions took place between February and June 2018. Note that the drop between June and July 2018 is artificial: for the purpose of this figure, all the women who gave birth during the roll out of the baseline were recoded as having given birth in July 2018 i.e. once information on the entire sample had been collected. The endline took place between February and June 2021. After January 2021, we cannot report the share of women who have not given birth in the whole sample because we do not observe all the women anymore. That is why the graph stops in January 2021. The first red vertical line indicates the first month when the whole sample could be affected in terms of births (9 months after the endline of the baseline). The second red line indicates 9 months after the start of the COVID-19 pandemic. The third red line indicates 9 months after the announcement of the national family planning policy (FPP). The first shaded area indicates 9 months after the period of the strike.